

Convolution and Correlation

2.1 INTRODUCTION

In the previous chapter we discussed a number of basic system properties. Of these, linearity and time invariance play a fundamental role in signal and system analysis because of the many physical phenomena that can be modelled by linear time-invariant (LTI) systems and because a mathematical analysis of the behaviour of such systems can be carried out in a fairly straightforward manner. In this chapter, we begin the development by deriving and examining a fundamental and extremely useful representation for LTI systems and by introducing an important class of these systems.

One of the primary reasons LTI systems are amenable to analysis is that any such system possesses the superposition property described in Section 1.11.1. As a consequence, if we can represent the input to an LTI system in terms of a linear combination of a set of basic signals, we can then use superposition to compute the output of the system in terms of its response to these basic signals.

A fundamental problem in system analysis is determining the response to some specified input. Analytically, this can be answered in many different ways. One obvious way is to solve the differential equation describing the system, subject to the specified input and initial conditions. In the following section, we introduce a second method that exploits the linearity and time invariance of the system. We begin by characterizing an LTI system in terms of its *impulse response*, defined as the output of an LTI system due to a unit impulse signal input applied at time $t = 0$ or $n = 0$. The impulse response completely characterizes the behaviour of any LTI system. Given the impulse response, we determine the output due to an arbitrary input signal by expressing the input as a weighted superposition of time-shifted impulses. For linearity and time invariance, the output signal must be a weighted superposition of time-shifted impulse responses. This weighted superposition is termed the *convolution sum* for discrete-time systems and the *convolution integral* for continuous-time systems.

2.2 CONTINUOUS-TIME LTI SYSTEMS: THE CONVOLUTION INTEGRAL

The goal of this section is to obtain a complete characterization of a continuous-time LTI system in terms of its unit impulse response.

2.2.1 Unit Impulse Response

The impulse response is defined as the output of an LTI system due to a unit impulse signal input applied at time $t = 0$ or $n = 0$ (Fig. 2.1).

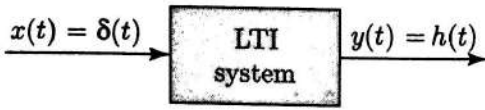


Fig. 2.1 Continuous-time LTI system

$$x(t) \rightarrow y(t)$$

$$\delta(t) \rightarrow h(t)$$

where $\delta(t)$ is the unit impulse function and $h(t)$ is the unit impulse response of a continuous-time LTI system (CTS). Similarly for a discrete-time system (DTS), $\delta(n)$ is the unit impulse or unit sample input and $h(n)$ is the unit impulse response or unit sample response (Fig. 2.2).

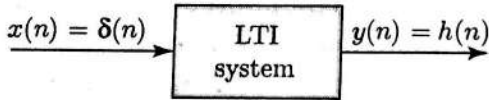


Fig. 2.2 Discrete-time LTI system

$$x(n) \rightarrow y(n)$$

$$\delta(n) \rightarrow h(n)$$

2.2.2 Convolution Integral

Linear systems are governed by the superposition principle. Let the responses of the system to two inputs $x_1(t)$ and $x_2(t)$ be $y_1(t)$ and $y_2(t)$, respectively. The system is linear if the response to the input $x(t) = a_1x_1(t) + a_2x_2(t)$ is equal to $y(t) = a_1y_1(t) + a_2y_2(t)$. More generally, if the input $x(t)$ is the weighted sum of any set signals $x_i(t)$, if the response to $x_i(t)$ is $y_i(t)$, and if the system is linear, the output $y(t)$ will be the weighted sum of the response $y_i(t)$. That is, if

$$x(t) = a_1x_1(t) + a_2x_2(t) + \cdots + a_Nx_N(t) = \sum_{i=1}^N a_i x_i(t)$$

we have

$$y(t) = a_1y_1(t) + a_2y_2(t) + \cdots + a_Ny_N(t) = \sum_{i=1}^N a_i y_i(t)$$

In Section 1.4.2 [Property 1, i.e., Eq. (1.10)], we demonstrated that the unit impulse function can be used as basic building blocks to represent arbitrary signals. In fact, the shifting property of the δ function,

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

shows that any signal $x(t)$ can be expressed as a continuum of weighted impulses.

Now consider a continuous-time LTI system with input $x(t)$. Using the superposition and time-invariant property of LTI systems, we can express output $y(t)$ as a linear combination of the responses of the system to shifted impulse signals.

$$\begin{aligned}
 x(t) &\longrightarrow y(t) \\
 \delta(t) &\longrightarrow h(t) \\
 \delta(t - \tau) &\longrightarrow h(t - \tau) \\
 x(\tau)\delta(t - \tau) &\longrightarrow x(\tau)h(t - \tau) \\
 x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t - \tau) d\tau &\longrightarrow y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau
 \end{aligned}$$

Thus, we obtain the system output $y(t)$ to an arbitrary input $x(t)$ in terms of the unit impulse response $h(t)$,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau \quad (2.1)$$

The integral relationship expressed in Eq. (2.1) is called the *convolution integral* of signals $x(t)$ and $h(t)$. This operation is represented symbolically as

$$y(t) = x(t) * h(t) \quad (2.2)$$

Continuous-convolution (Convolution Integral) Evaluation

It is useful to present examples of the evaluation of continuous convolution before indicating its properties. Our first example considers two functions, each defined by a single equation for all values of the independent variable. The convolution integral is conceptually very easy to evaluate because it consists of a single direct evaluation of the integral and produces a single equation for all values of the independent variable.

Example 2.1 Find the convolution of the two continuous-time signals

$$f(t) = e^{-t^2} \quad \text{and} \quad g(t) = 3t^2, \quad \text{for all } t$$

Solution

By definition

$$\begin{aligned}
 y(t) = f(t) * g(t) &= \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau \\
 &= \int_{-\infty}^{\infty} e^{-\tau^2} [3(t - \tau)^2] d\tau
 \end{aligned}$$

$$\begin{aligned}
&= 3t^2 \int_{-\infty}^{\infty} e^{-\tau^2} d\tau - 6t \int_{-\infty}^{\infty} \tau e^{-\tau^2} d\tau + 3 \int_{-\infty}^{\infty} \tau^2 e^{-\tau^2} d\tau \\
&= 3t^2 \sqrt{\pi} - 0 + 3 \frac{\sqrt{\pi}}{2} \\
&= 5.31t^2 + 2.659, \quad \text{for all } t
\end{aligned}$$

Next we will consider the convolution of a piecewise-defined function and a function defined by a single equation for all values of the independent variable. In this case the integral must be evaluated in two or more parts. However, the resulting convolution is a function defined by a single equation; hence, the integral needs to be evaluated only once to yield a result that is valid for all values of the independent variable.

Example 2.2 Find the convolution of the two continuous-time signals

$$f(t) = 3\cos 2t, \quad \text{for all } t \quad \text{and} \quad g(t) = e^{-|t|} = \begin{cases} e^t, & t < 0 \\ e^{-t}, & t \geq 0 \end{cases}$$

Solution

By definition

$$y(t) = f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau$$

Now

$$f(\tau) = 3\cos 2\tau, \quad \text{for all } \tau \quad \text{and} \quad g(t - \tau) = \begin{cases} e^{(t-\tau)}, & (t - \tau) < 0 \rightarrow \tau > t \\ e^{-(t-\tau)}, & (t - \tau) \geq 0 \rightarrow \tau \leq t \end{cases}$$

Therefore,

$$\begin{aligned}
y(t) &= \int_{-\infty}^t (3\cos 2\tau)(e^{(\tau-t)}) d\tau + \int_t^{\infty} (3\cos 2\tau)(e^{(t-\tau)}) d\tau \\
&= 3e^{-t} \int_{-\infty}^t e^{\tau} \cos 2\tau d\tau + 3e^t \int_t^{\infty} e^{-\tau} \cos 2\tau d\tau \\
&= 3e^{-t} \left[\frac{e^{\tau}(\cos 2\tau + 2\sin 2\tau)}{5} \right]_{-\infty}^t + 3e^t \left[\frac{e^{-\tau}(-\cos 2\tau + 2\sin 2\tau)}{5} \right]_t^{\infty} \\
&= \frac{6}{5} \cos 2t, \quad \text{for all } t
\end{aligned}$$

The convolution of two functions that are piecewise defined produces a convolution integral that must be evaluated in two or more parts. Since the resulting convolution is also a piecewise-defined function, the convolution integral must be evaluated as many times as there are pieces in the resulting function to find the mathematical expressions for those pieces. For the evaluation of the convolution of two piecewise-defined functions, it is helpful to express carefully the functions being multiplied to form the integrand and to sketch these functions in positions corresponding to each of the integral evaluations that must be made. We list in what follows the steps in evaluating the convolution defined by Eq. (2.1):

Step 1. Write equations and draw sketches for $x(t)$ and $h(t)$.

Step 2. Write equations and draw sketches for $x(\tau)$ and $h(t - \tau)$.

Step 3. Perform integrations to find $y(t)$ for all t .

The following examples illustrate the use of the evaluation steps.

Example 2.3 Find the convolution of the two continuous-time signals

$$x(t) = e^{-|t|}, \text{ for all } t \quad \text{and} \quad h(t) = \begin{cases} e^{-2t}, & t \geq 1 \\ 0, & t < 1 \end{cases}$$

Solution

By definition

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau$$

Step 1: Write the equations and draw sketches for $x(t)$ and $h(t)$ as shown in Fig. 2.3.

$$x(t) = \begin{cases} e^{-t}, & t \geq 0 \\ e^t, & t < 0 \end{cases} \quad h(t) = \begin{cases} e^{-2t}, & t \geq 1 \\ 0, & t < 1 \end{cases}$$

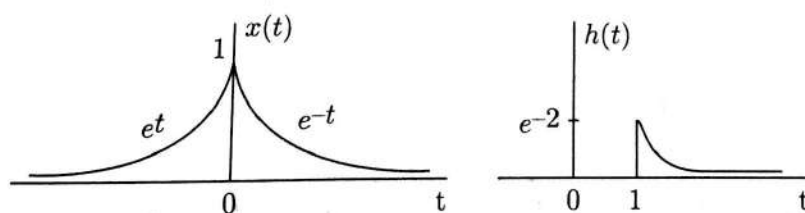


Fig. 2.3 Step 1 for Example 2.3

Step 2: Write the equations and draw sketches for $x(\tau)$ and $h(t - \tau)$ [apply precedence rule: $h(t) \rightarrow h(\tau) \rightarrow h(\tau + t) \rightarrow h(-\tau + t) = h(t - \tau)$] as shown in Fig. 2.4.

$$x(\tau) = \begin{cases} e^{-\tau}, & \tau \geq 0 \\ e^{\tau}, & \tau < 0 \end{cases} \quad h(t - \tau) = \begin{cases} e^{-2(t-\tau)}, & (t - \tau) \geq 1 \rightarrow \tau \leq (t - 1) \\ 0, & (t - \tau) < 1 \rightarrow \tau > (t - 1) \end{cases}$$

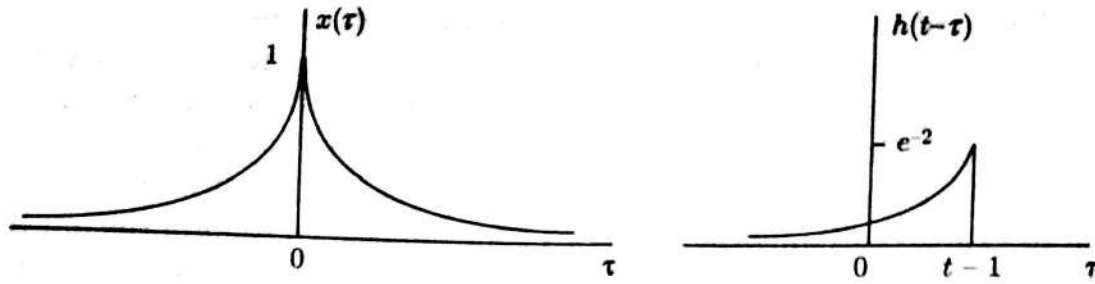


Fig. 2.4 Step 2 for Example 2.3

Step 3: Perform integration to find $y(t)$ for all values of t .

Case I: $(t-1) < 0 \rightarrow t < 1$. Figure 2.5 shows $x(\tau)$ and $h(t-\tau)$. The two functions overlap only over the range $-\infty \leq \tau \leq t-1$ (shaded interval) such that

$$y(t) = \int_{-\infty}^{t-1} e^{\tau} e^{-2(t-\tau)} d\tau + \int_{t-1}^{\infty} x(\tau) \times 0 d\tau = e^{-2t} \int_{-\infty}^{t-1} e^{3\tau} d\tau + 0 = \frac{1}{3} e^{(t-3)} \quad t < 1$$

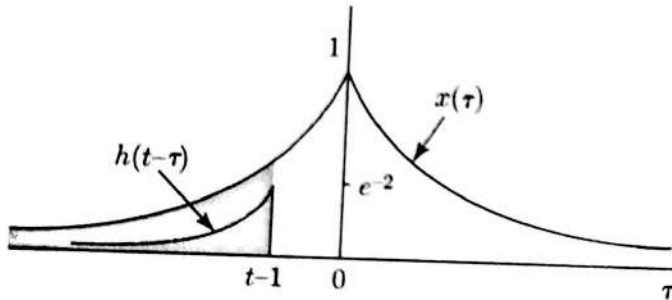


Fig. 2.5 Signal sketch for step 3, case I

Case II: $(t-1) \geq 0 \rightarrow t \geq 1$. Figure 2.6 shows $x(\tau)$ and $h(t-\tau)$. The two functions overlap only over the range $-\infty \leq \tau \leq t-1$ (shaded interval) such that

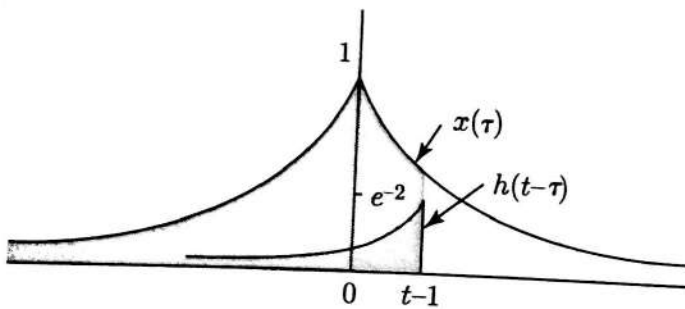


Fig. 2.6 Signal sketch for step 3, case II

$$\begin{aligned} y(t) &= \int_{-\infty}^0 e^{\tau} e^{-2(t-\tau)} d\tau + \int_0^{t-1} e^{-\tau} e^{-2(t-\tau)} d\tau + \int_{t-1}^{\infty} e^{-\tau} 0 d\tau \\ &= e^{-2t} \left(\int_{-\infty}^0 e^{3\tau} d\tau + \int_0^{t-1} e^{\tau} d\tau \right) \\ &= e^{-(t+1)} - \frac{2}{3} e^{-2t}, \quad t \geq 1 \end{aligned}$$

In summary,

$$y(t) = \begin{cases} e^{-(t+1)} - \frac{2}{3}e^{-2t}, & t \geq 1 \\ \frac{1}{3}e^{(t-3)}, & t < 1 \end{cases}$$

The solution is illustrated in Fig. 2.7.

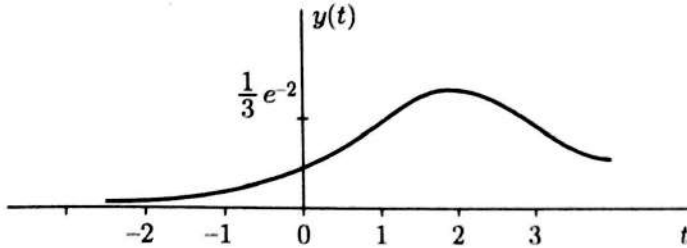


Fig. 2.7 Convolution result

Example 2.4 Find the convolution of a rectangle signal (or gate function) $x(t)$ with itself.

$$x(t) = A \operatorname{rect}\left(\frac{t}{2T}\right) = A \Pi\left(\frac{t}{2T}\right) = \begin{cases} A, & -T < t < T \\ 0, & \text{otherwise} \end{cases}$$

Or, evaluate the convolution integral for a system with input $x(t)$ and impulse response $h(t)$, respectively, given by

$$x(t) = h(t) = A[u(t+T) - u(t-T)]$$

Solution

By definition

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau$$

Step 1: Write the equations and draw sketches for $x(t)$ and $h(t)$ as shown in Fig. 2.8.

$$x(t) = \begin{cases} A, & -T < t < T \\ 0, & \text{otherwise} \end{cases} \quad h(t) = \begin{cases} A, & -T < t < T \\ 0, & \text{otherwise} \end{cases}$$

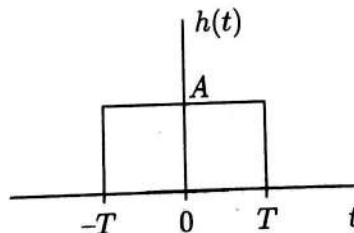
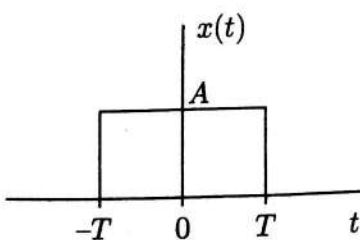


Fig. 2.8 Step 1 for Example 2.4

Step 2: Write the equations and draw sketches for $x(\tau)$ and $h(t - \tau)$ [apply precedence rule: $h(t) \rightarrow h(\tau) \rightarrow h(\tau + t) \rightarrow h(-\tau + t) = h(t - \tau)$] as shown in Fig. 2.9.

$$x(\tau) = \begin{cases} A, & -T < \tau < T \\ 0, & \text{otherwise} \end{cases} \quad h(t - \tau) = \begin{cases} A, & -T < (t - \tau) < T \longrightarrow (t - T) < \tau < (t + T) \\ 0, & \text{otherwise} \end{cases}$$

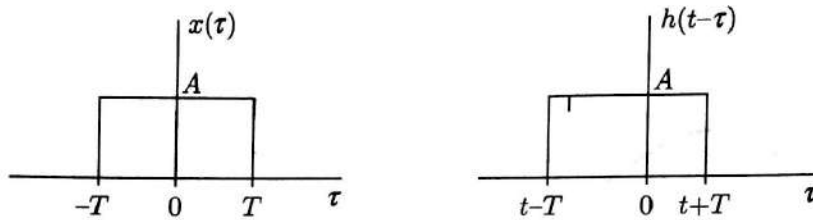


Fig. 2.9 Step 2 for Example 2.4

Step 3: Perform integration to find $y(t)$ for all values of t .

Case I: $t + T < -T \longrightarrow t < -2T$. Figure 2.10 shows $x(\tau)$ and $h(t - \tau)$. We see that the signals do not overlap (Fig. 2.10); hence, the product of the integrand is zero. Thus,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau = 0$$

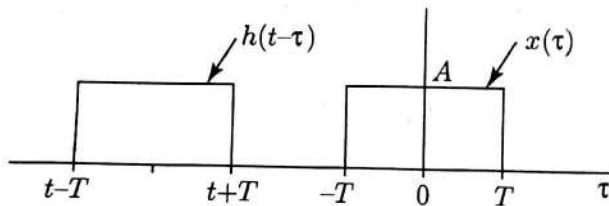


Fig. 2.10 Signal sketch for step 3, case I

Case II: $-T \leq (t + T) < T \longrightarrow -2T \leq t < 0$. Figure 2.11 shows $x(\tau)$ and $h(t - \tau)$. The two functions overlap only over the interval $-T \leq \tau \leq (t + T)$ (shaded interval) such that

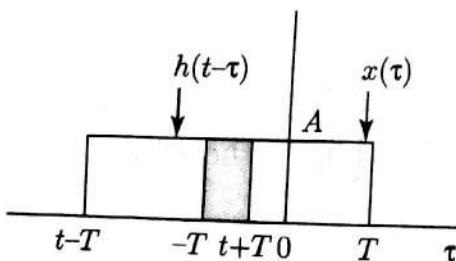


Fig. 2.11 Signal sketch for step 3, case II

$$y(t) = \int_{-T}^{t+T} A \cdot A d\tau = A^2(t + 2T), \quad -2T \leq t < 0$$

Case III: $-T \leq t - T \leq T \rightarrow 0 \leq t \leq 2T$. Figure 2.12 shows $x(\tau)$ and $h(t - \tau)$. The two functions overlap only over the interval $(t - T) \leq \tau \leq T$ (shaded interval) such that

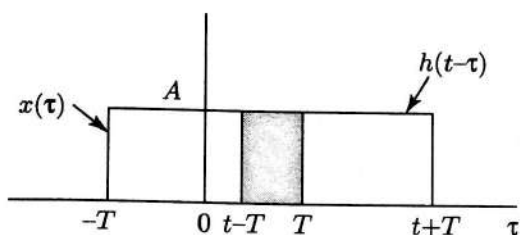


Fig. 2.12 Signal sketch for step 3, case III

$$y(t) = \int_{t-T}^T A \cdot A d\tau = A^2(2T - t), \quad 0 \leq t \leq 2T$$

Case IV: $t - T > T \rightarrow t > 2T$. Figure 2.13 shows $x(\tau)$ and $h(t - \tau)$. The two functions do not overlap (Fig. 2.13); hence, the product of the integrand is zero.

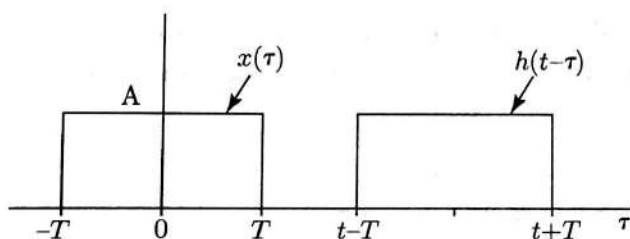


Fig. 2.13 Signal sketch for step 3, case IV

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau = 0, \quad t > 2T$$

In summary

$$y(t) = 2TA^2 \Delta\left(\frac{t}{2T}\right) = \begin{cases} 0, & t < -2T \\ A^2(t + 2T), & -2T \leq t < 0 \\ A^2(2T - t), & 0 \leq t \leq 2T \\ 0, & t > 2T \end{cases}$$

or, in more compact form

$$y(t) = \begin{cases} A^2(2T - |t|), & |t| \leq 2T \\ 0, & |t| \geq 2T \end{cases}$$

This signal (see Fig. 2.14) is called the triangular signal. We use the notation $\Delta(t/2T)$ to denote the triangular signal that is of unit height, centred around $t = 0$, and has base of length $4T$.

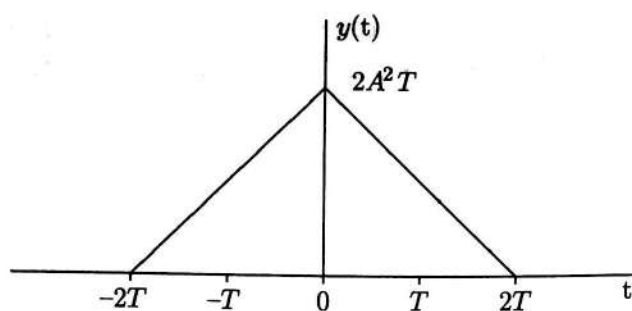


Fig. 2.14 Convolution result

2.3 PROPERTIES OF CONVOLUTION INTEGRAL

Continuous-time convolution satisfies the following properties:

2.3.1 Commutative Property

The convolution operation possesses the commutative property, or

$$x(t) * h(t) = h(t) * x(t) \quad (2.3)$$

This property (see Fig. 2.15), implies that the role of the input signal and impulse response are interchangeable. It is often used to simplify the evaluation or interpretation of the convolution integral.



Fig. 2.15

Proof By definition

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

If we perform a change of variables by letting $(t - \tau) = \alpha$, then $\tau = (t - \alpha)$, $d\alpha = -d\tau$, $\alpha \rightarrow \infty$ as $\tau \rightarrow -\infty$, and $\alpha \rightarrow -\infty$ as $\tau \rightarrow \infty$. Therefore,

$$x(t) * h(t) = - \int_{\infty}^{-\infty} x(t - \alpha) h(\alpha) d\alpha = \int_{-\infty}^{\infty} h(\alpha) x(t - \alpha) d\alpha = h(t) * x(t) \quad \blacksquare$$

2.3.2 Associative Property

The continuous convolution is associative, i.e.,

$$x(t) * [h_1(t) * h_2(t)] = [x(t) * h_1(t)] * h_2(t) \quad (2.4)$$

Associativity, as shown in Fig. 2.16, implies that a cascade combination of LTI systems can be replaced by a single system whose impulse response is the convolution of the individual impulse responses.

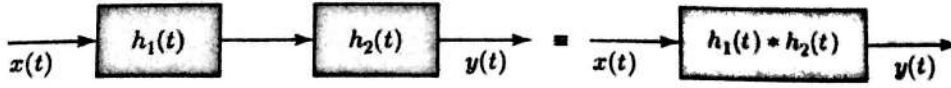


Fig. 2.16

Proof By definition

$$\begin{aligned} x(t) * [h_1(t) * h_2(t)] &= x(t) * \left[\int_{-\infty}^{\infty} h_1(\tau) h_2(t - \tau) d\tau \right] \\ &= \int_{-\infty}^{\infty} x(\alpha) \left[\int_{-\infty}^{\infty} h_1(\tau) h_2(t - \alpha - \tau) d\tau \right] d\alpha \end{aligned}$$

A change of variables is performed by letting $\tau = \beta - \alpha$, which also yields $d\tau = d\beta$, $\beta \rightarrow \infty$ as $\tau \rightarrow \infty$, and $\beta \rightarrow -\infty$ as $\tau \rightarrow -\infty$. Therefore,

$$x(t) * [h_1(t) * h_2(t)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(\alpha) h_1(\beta - \alpha) h_2(t - \beta) d\beta d\alpha$$

Interchanging the order of integration gives the desired result

$$\begin{aligned} x(t) * [h_1(t) * h_2(t)] &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\alpha) h_1(\beta - \alpha) d\alpha \right] h_2(t - \beta) d\beta \\ &= \int_{-\infty}^{\infty} [x(\beta) * h_1(\beta)] h_2(t - \beta) d\beta = [x(t) * h_1(t)] * h_2(t) \end{aligned}$$

Since continuous convolution is associative, there is no need to indicate which convolution is to be performed first; hence, the square brackets can be removed from Eq. (2.4). Since continuous convolution is also commutative, then

$$x(t) * h_1(t) * h_2(t) = x(t) * h_2(t) * h_1(t) = \cdots = h_2(t) * h_1(t) * x(t) \quad \blacksquare$$

2.3.3 Distributive Property

The convolution operation possesses the distributive property, or

$$x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + x(t) * h_2(t) \quad (2.5)$$

Distributivity, as shown in Fig. 2.17, states that a parallel combination of LTI systems is equivalent to a single system whose impulse response is the sum of the individual impulse responses in the parallel configuration.

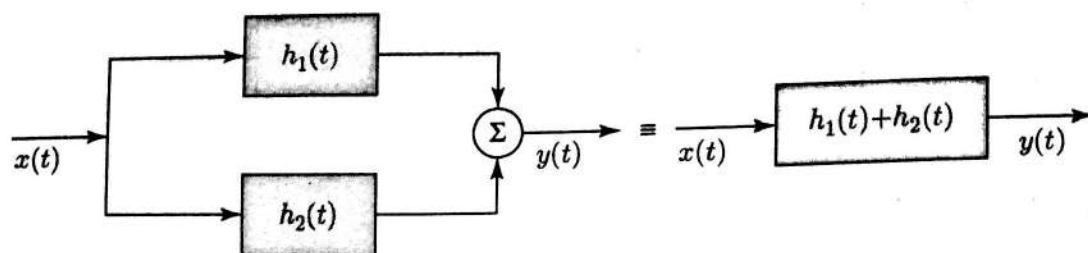


Fig. 2.17

• **Proof** By definition

$$\begin{aligned}
 x(t) * [h_1(t) + h_2(t)] &= \int_{-\infty}^{\infty} x(\tau) [h_1(t - \tau) + h_2(t - \tau)] d\tau \\
 &= \int_{-\infty}^{\infty} x(\tau) h_1(t - \tau) d\tau + \int_{-\infty}^{\infty} x(\tau) h_2(t - \tau) d\tau \\
 &= [x(t) * h_1(t)] + [x(t) * h_2(t)]
 \end{aligned}$$

2.3.4 Shift Property

According to the shift property,

$$\text{if } x(t) * h(t) = y(t), \quad \text{then } x(t) * h(t - t_0) = x(t - t_0) * h(t) = y(t - t_0) \quad (2.6)$$

and

$$x(t - t_1) * h(t - t_2) = y(t - t_1 - t_2) \quad (2.7)$$

Proof By definition

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = y(t)$$

$$x(t) * h(t - t_0) = \int_{-\infty}^{\infty} x(\tau) h(t - t_0 - \tau) d\tau = y(t - t_0)$$

Equation (2.7) follows from Eq. (2.6).

2.3.5 Convolution with an Impulse

The convolution of a signal $x(t)$ with a unit impulse function results in the signal $x(t)$ itself

$$x(t) * \delta(t) = x(t) \quad (2.8)$$

Proof By definition

$$x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

From the sampling property of the impulse function, we have

$$x(\tau)\delta(t - \tau) = x(t)\delta(t - \tau)$$

and therefore

$$x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t - \tau) d\tau = x(t) \int_{-\infty}^{\infty} \delta(t - \tau) d\tau = x(t)$$

2.3.6 Width Property

The width of the nonzero extent (the interval of time between the first and last nonzero values) of the continuous convolution of two functions equals the sum of the widths of the nonzero extents of the two functions. In other words, if the durations (widths) of $x(t)$ and $h(t)$ are finite, given by W_x and W_h , respectively, then the duration (width) of $x(t) * h(t)$ is $W_x + W_h$.

Example 2.5 Two continuous-time signals $x(t)$ and $h(t)$ are shown in Fig. 2.18. Where will the right and left edges of the convolution of $x(t)$ and $h(t)$ be located? What will be the width of the nonzero extent of the convolution?

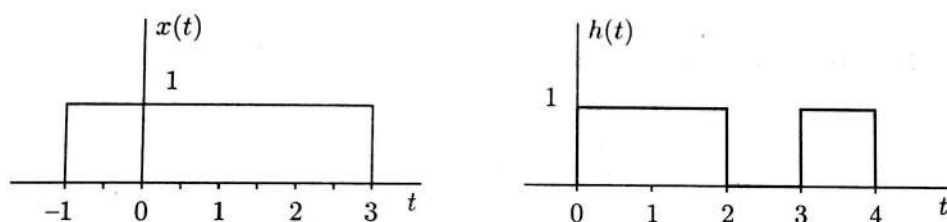


Fig. 2.18 Signal $x(t)$ and $h(t)$

Solution

$$\text{Right edge of } x(t) = \text{RE}_x = 3$$

$$\text{Right edge of } h(t) = \text{RE}_h = 4$$

$$\text{Left edge of } x(t) = \text{LE}_x = -1$$

$$\text{Left edge of } h(t) = \text{LE}_h = 0$$

$$\text{Duration (width) of } x(t) = W_x = \text{RE}_x - \text{LE}_x = 3 - (-1) = 4$$

$$\text{Duration (width) of } h(t) = W_h = \text{RE}_h - \text{LE}_h = 4 - 0 = 4$$

$$y(t) = x(t) * h(t)$$

$$\text{Right edge of } y(t) = \text{RE}_y = \text{RE}_x + \text{RE}_h = 3 + 4 = 7$$

$$\text{Left edge of } y(t) = \text{LE}_y = \text{LE}_x + \text{LE}_h = (-1) + 0 = -1$$

$$\text{Duration (width) of } y(t) = W_y = W_x + W_h = 4 + 4 = 8.$$

The result of the convolution of $x(t)$ and $h(t)$ is shown in Fig. 2.19. One can verify that it is correct.

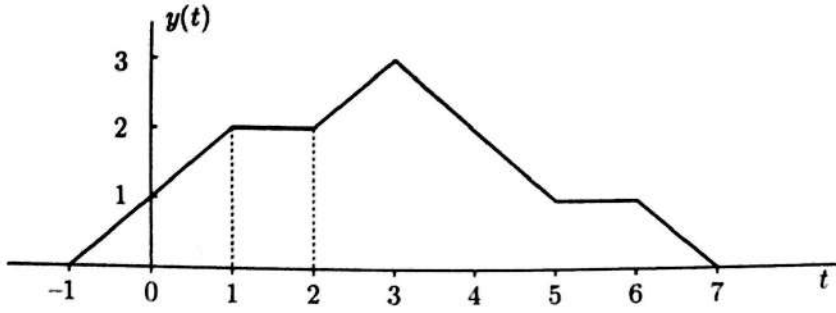


Fig. 2.19 Convolution result

2.3.7 Differentiation Property

According to the differentiation property,

$$\text{if } x(t) * h(t) = y(t), \quad \text{then } \left(\frac{d}{dt} x(t) \right) * h(t) = x(t) * \left(\frac{d}{dt} h(t) \right) = \frac{d}{dt} y(t) \quad (2.9)$$

Proof By definition

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

By differentiating both sides of the above equation w.r.t. t , we get

$$\begin{aligned} \frac{d}{dt} y(t) &= \int_{-\infty}^{\infty} x(\tau) \left(\frac{d}{dt} h(t - \tau) \right) d\tau \\ &= x(t) * \left(\frac{d}{dt} h(t) \right) \end{aligned}$$

2.3.8 Time-scaling Property

According to the time-scaling property,

$$\text{if } x(t) * h(t) = y(t), \quad \text{then } x(at) * h(at) = \frac{1}{|a|} y(at) \quad (2.10)$$

This property states that if both $x(t)$ and $h(t)$ are time-scaled by a , their convolution is also time-scaled by a and multiplied by $1/|a|$.

Proof Case I: $a > 0$. By definition

$$\begin{aligned} x(t) * h(t) &= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = y(t) \\ x(at) * h(at) &= \int_{-\infty}^{\infty} x(a\tau) h(a(t - \tau)) d\tau = \int_{-\infty}^{\infty} x(a\tau) h(at - a\tau) d\tau \end{aligned}$$

A change of variables is performed by letting $a\tau = \alpha$, which also yields $d\tau = (1/a)d\alpha$, $\alpha \rightarrow \infty$ as $\tau \rightarrow \infty$, and $\alpha \rightarrow -\infty$ as $\tau \rightarrow -\infty$. Therefore,

$$x(at) * h(at) = \frac{1}{a} \int_{-\infty}^{\infty} x(\alpha)h(at - \alpha) d\alpha = \frac{1}{a}y(at)$$

Case II: $a < 0$. By definition

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau = y(t)$$

$$x(-at) * h(-at) = \int_{-\infty}^{\infty} x(-a\tau)h(-a(t - \tau)) d\tau = \int_{-\infty}^{\infty} x(-a\tau)h(-at + a\tau) d\tau$$

A change of variables is performed by letting $-a\tau = \alpha$, which also yields $d\tau = -(1/a)d\alpha$, $\alpha \rightarrow \infty$ as $\tau \rightarrow -\infty$, and $\alpha \rightarrow -\infty$ as $\tau \rightarrow \infty$. Therefore,

$$x(at) * h(at) = \frac{1}{a} \int_{-\infty}^{\infty} x(\alpha)h(at - \alpha) d\alpha = \frac{1}{a}y(at)$$

From the above two cases it is evident that

$$x(at) * h(at) = \frac{1}{|a|}y(at)$$

Example 2.6 Show that

- (a) the convolution of an odd and an even function is an odd function.
- (b) the convolution of two odd functions is an even function.
- (c) the convolution of two even functions is an even function.

Solution

(a) Let $x(t)$ and $h(t)$ be an odd and even function, respectively, i.e., $x(t) = -x(-t)$ and $h(t) = h(-t)$. Using time-scaling property, we have

$$x(t) * h(t) = y(t)$$

$$x(at) * h(at) = \frac{1}{|a|}y(at)$$

Substituting $a = -1$ into the above equation, we get

$$x(-t) * h(-t) = y(-t)$$

$$-x(t) * h(t) = y(-t)$$

$$x(t) * h(t) = -y(-t) = y(t) \quad \text{an odd function}$$

(b) Let both $x(t)$ and $h(t)$ be odd functions, i.e., $x(t) = -x(-t)$ and $h(t) = -h(-t)$. Using time-scaling property, we have

$$x(t) * h(t) = y(t)$$

$$x(at) * h(at) = \frac{1}{|a|} y(at)$$

Substituting $a = -1$ into the above equation, we get

$$x(-t) * h(-t) = y(-t)$$

$$-x(t) * -h(t) = y(-t)$$

$$x(t) * h(t) = y(-t) = y(t) \quad \text{an even function}$$

(c) Let both $x(t)$ and $h(t)$ be even functions, i.e., $x(t) = x(-t)$ and $h(t) = h(-t)$. Using time-scaling property, we have

$$x(t) * h(t) = y(t)$$

$$x(at) * h(at) = \frac{1}{|a|} y(at)$$

Substituting $a = -1$ into the above equation, we get

$$x(-t) * h(-t) = y(-t)$$

$$x(t) * h(t) = y(-t) = y(t) \quad \text{an even function}$$

Example 2.7 The convolution has the property that the area of the convolution integral is equal to the product of the two signals entering into the convolution. We define the area under a continuous-time signal $y(t)$ as

$$A_y = \int_{-\infty}^{\infty} y(t) dt$$

Show that if $y(t) = x(t) * h(t)$, then $A_y = A_x A_h$

Solution

By definition

$$y(t) = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

$$\int_{-\infty}^{\infty} y(t) dt = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \right] dt$$

Interchanging the orders of the integration results in

$$\int_{-\infty}^{\infty} y(t) dt = \int_{-\infty}^{\infty} x(\tau) \underbrace{\left[\int_{-\infty}^{\infty} h(t - \tau) dt \right]}_{A_h} d\tau$$

$$A_y = \int_{-\infty}^{\infty} x(\tau) A_h d\tau = \int_{-\infty}^{\infty} x(\tau) d\tau A_h = A_x A_h$$

Example 2.8 Prove the following convolution integrals:

(a) $x(t) * \delta(t - t_0) = x(t - t_0)$

(b) $x(t) * \delta(t + t_0) = x(t + t_0)$

(c) $x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$

(d) $x(t) * u_1(t) = x(t) * \frac{d\delta(t)}{dt} = \frac{dx(t)}{dt}$

(e) $u(t) * u(t) = tu(t) = r(t)$

Solution

(a) By definition

$$x(t) * \delta(t - t_0) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau - t_0) d\tau$$

From the sampling property of the impulse function, we have

$$x(\tau) \delta(t - \tau - t_0) = x(t - t_0) \delta(t - \tau - t_0)$$

and therefore

$$x(t) * \delta(t - t_0) = \int_{-\infty}^{\infty} x(t - t_0) \delta(t - \tau - t_0) d\tau = x(t - t_0) \underbrace{\int_{-\infty}^{\infty} \delta(t - \tau - t_0) d\tau}_{=1} = x(t - t_0)$$

(b) By definition

$$x(t) * \delta(t + t_0) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau + t_0) d\tau$$

From the sampling property of the impulse function, we have

$$x(\tau) \delta(t - \tau + t_0) = x(t + t_0) \delta(t - \tau + t_0)$$

and therefore

$$x(t) * \delta(t + t_0) = \int_{-\infty}^{\infty} x(t + t_0) \delta(t - \tau + t_0) d\tau = x(t + t_0) \underbrace{\int_{-\infty}^{\infty} \delta(t - \tau + t_0) d\tau}_{=1} = x(t + t_0)$$

(c) By definition

$$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau) u(t - \tau) d\tau$$

Since

$$u(t - \tau) = \begin{cases} 1, & t - \tau > 0 \longrightarrow \tau < t \\ 0, & \text{otherwise} \end{cases}$$

Therefore, we have

$$x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$$

(d) By definition

$$\begin{aligned} x(t) * u_1(t) &= x(t) * \frac{d\delta(t)}{dt} = x(t) * \delta'(t) = \int_{-\infty}^{\infty} x(\tau) \delta'(t - \tau) d\tau \\ &= - \int_{-\infty}^{\infty} x'(\tau) [-\delta(t - \tau)] d\tau \\ &= \int_{-\infty}^{\infty} x'(\tau) \delta(t - \tau) d\tau \\ &= x'(\tau) \Big|_{\tau=t} = x'(t) \end{aligned}$$

(e) By definition

$$u(t) * u(t) = \int_{-\infty}^{\infty} u(\tau) u(t - \tau) d\tau$$

Since

$$u(t - \tau) = \begin{cases} 1, & t - \tau > 0 \longrightarrow \tau < t \\ 0, & \text{otherwise} \end{cases}$$

Therefore, we have

$$u(t) * u(t) = \int_{-\infty}^t u(\tau) d\tau = \int_0^t (1) d\tau = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases} = r(t)$$

Example 2.9 Determine and sketch the convolution of the following two signals:

$$x(t) = \begin{cases} 1, & -1 < t < 1 \\ 0, & \text{elsewhere} \end{cases} \quad \text{and} \quad h(t) = \delta(t+1) + 2\delta(t+2)$$

Solution

Using the result of part (b) of Example 2.8, we have

$$\begin{aligned} y(t) &= x(t) * h(t) = x(t) * [\delta(t+1) + 2\delta(t+2)] \\ &= [x(t) * \delta(t+1)] + [x(t) * 2\delta(t+2)] \\ &= x(t+1) + 2x(t+2) \end{aligned}$$

The signals $x(t)$, $h(t)$, and $y(t)$ are shown in Fig. 2.20.

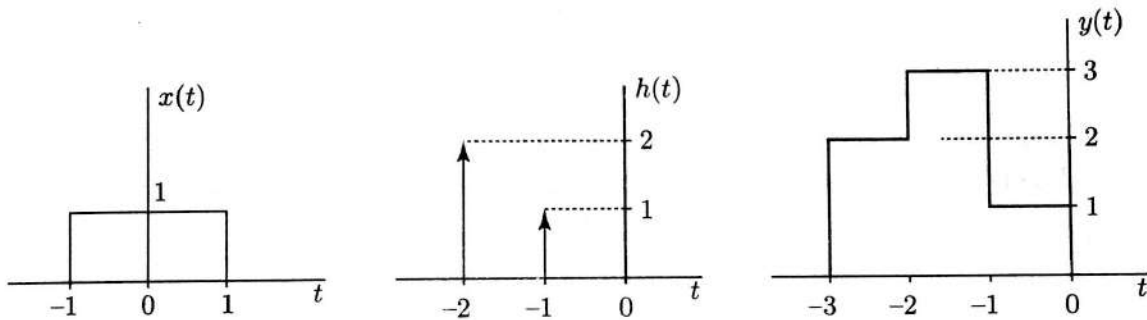


Fig. 2.20

Example 2.10 Determine and sketch the convolution of the following two signals:

$$x(t) = \begin{cases} t+1, & 0 \leq t \leq 1 \\ 2-t, & 1 < t \leq 2 \\ 0, & \text{elsewhere} \end{cases} \quad \text{and} \quad h(t) = \delta(t+2) + 2\delta(t+1)$$

Solution

Using the result of part (b) of Example 2.8, we have

$$\begin{aligned} y(t) &= x(t) * h(t) = x(t) * [\delta(t+2) + 2\delta(t+1)] \\ &= x(t) * \delta(t+2) + x(t) * 2\delta(t+1) \\ y(t) &= x(t+2) + 2x(t+1) \end{aligned}$$

The signals $x(t)$, $h(t)$, and $y(t)$ are shown in Fig. 2.21. Using these plots, we can easily show that

$$y(t) = \begin{cases} t+3, & -2 < t \leq 1 \\ t+4, & -1 < t \leq 0 \\ 2-2t, & 0 < t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

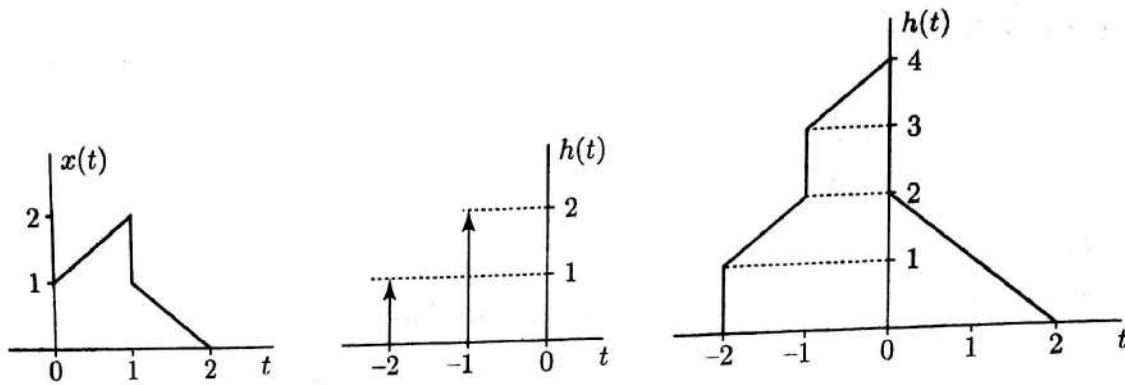


Fig. 2.21

Example 2.11 Suppose that

$$x(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

and $h(t) = x(t/\alpha)$, where $0 < \alpha \leq 1$.

(a) Determine and sketch $y(t) = x(t) * h(t)$.

(b) If $dy(t)/dt$ contains only three discontinuities, what is the value of α ?

Solution

By definition

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau$$

Step 1: Write the equations and draw sketches for $x(t)$ and $h(t)$ as shown in Fig. 2.22.

$$x(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & \text{elsewhere} \end{cases} \quad h(t) = x\left(\frac{t}{\alpha}\right) = \begin{cases} 1, & 0 \leq t/\alpha \leq 1 \rightarrow 0 \leq t \leq \alpha \\ 0, & \text{elsewhere} \end{cases}$$

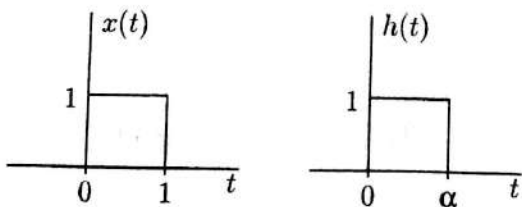


Fig. 2.22 Step 1 for Example 2.11

Step 2: Write the equations and draw sketches for $x(\tau)$ and $h(t - \tau)$ [apply precedence rule: $h(t) \rightarrow h(\tau) \rightarrow h(\tau + t) \rightarrow h(-\tau + t) = h(t - \tau)$] as shown in Fig. 2.23.

$$x(\tau) = \begin{cases} 1, & 0 \leq \tau \leq 1 \\ 0, & \text{elsewhere} \end{cases} \quad h(t - \tau) = \begin{cases} 1, & 0 \leq (t - \tau) \leq \alpha \rightarrow (t - \alpha) \leq \tau \leq t \\ 0, & \text{elsewhere} \end{cases}$$

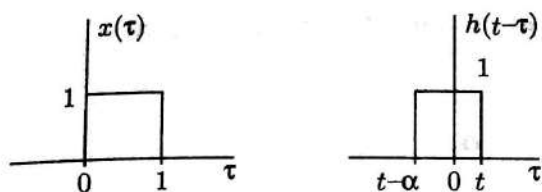


Fig. 2.23 Step 2 for Example 2.11

Step 3: Perform integration to find $y(t)$ for all values of t .

Case I: $t < 0$. Figure 2.24 shows $x(\tau)$ and $h(t - \tau)$.

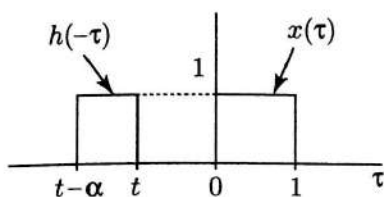


Fig. 2.24 Signal sketch for step 3, case I

We see that the signals do not overlap (Fig. 2.24); hence, the product of the integrand is zero. Thus,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau = 0$$

Case II: $t \geq 0$ and $(t - \alpha) < 0 \rightarrow 0 \leq t < \alpha$. Figure 2.25 shows $x(\tau)$ and $h(t - \tau)$.

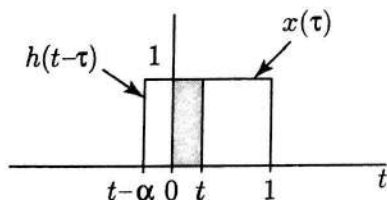


Fig. 2.25 Signal sketch for step 3, case II

The two signals overlap over the interval $0 \leq \tau \leq t$ (shaded interval) such that

$$y(t) = \int_0^t 1 d\tau = t, \quad 0 \leq t < \alpha$$

Case III: $(t - \alpha) \geq 0$ and $t \leq 1 \rightarrow \alpha \leq t \leq 1$. Figure 2.26 shows $x(\tau)$ and $h(t - \tau)$.

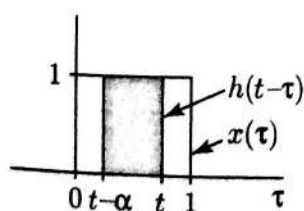


Fig. 2.26 Signal sketch for step 3, case III

The two signals overlap over the interval $(t - \alpha) \leq \tau \leq t$ such that

$$y(t) = \int_{t-\alpha}^t 1 d\tau = \alpha, \quad \alpha \leq t \leq 1$$

Case IV: $t - \alpha \leq 1$ and $t > 1 \rightarrow 1 < t \leq (\alpha + 1)$. Figure 2.27 shows $x(\tau)$ and $h(t - \tau)$.

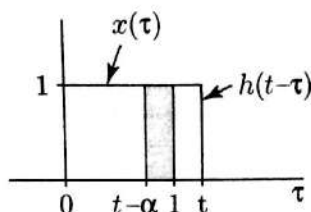


Fig. 2.27 Signal sketch for step 3, case IV

The two signals overlap over the interval $(t - \alpha) \leq \tau \leq 1$ such that

$$y(t) = \int_{t-\alpha}^1 1 d\tau = 1 - t + \alpha, \quad 1 < t \leq (\alpha + 1)$$

Case V: $(t - \alpha) > 1 \rightarrow t > (\alpha + 1)$. Figure 2.28 shows $x(\tau)$ and $h(t - \tau)$.

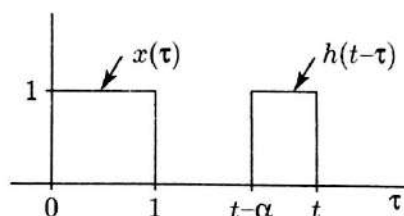


Fig. 2.28 Signal sketch for step 3, case V

We see that the signals do not overlap (Fig. 2.28); hence, the product of the integrand is zero. Thus,

$$y(t) = 0, \quad t > (\alpha + 1)$$

In summary

$$y(t) = \begin{cases} 0, & t < 0 \\ t, & 0 \leq t < \alpha \\ \alpha, & \alpha \leq t < 1 \\ 1 - t + \alpha, & 1 \leq t \leq (\alpha + 1) \\ 0, & t > (\alpha + 1) \end{cases}$$

The signal $y(t)$ is shown in Fig. 2.29.

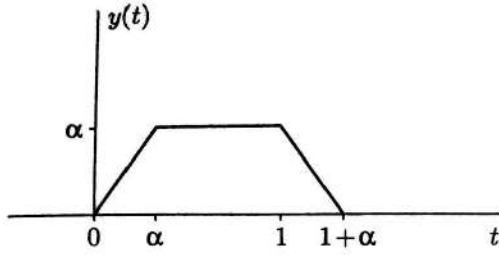
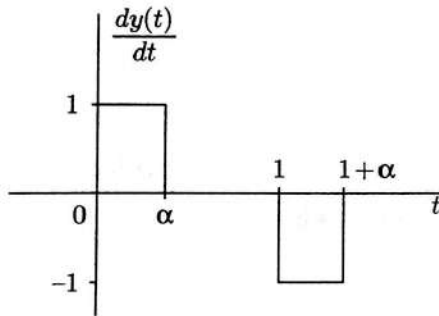


Fig. 2.29 Convolution result

(b) The derivative $(dy(t)/dt)$ of $y(t)$ is shown in Fig. 2.30. From the plot of $(dy(t)/dt)$ it is clear that it has discontinuities at 0, α , 1, and $(1 + \alpha)$. If we want only three discontinuities, then we need to ensure that $\alpha = 1$.

Fig. 2.30 Derivative of $y(t)$

Example 2.12 Consider an LTI system with input and output related through the following equation:

$$y(t) = \int_{-\infty}^t e^{-(t-\tau)} x(\tau - 2) d\tau$$

What is the impulse response $h(t)$ for this system?

Solution

Given that

$$y(t) = \int_{-\infty}^t e^{-(t-\tau)} x(\tau - 2) d\tau$$

A change of variables is performed by letting $\alpha = (\tau - 2)$, which also yields $d\alpha = d\tau$, $\alpha \rightarrow -\infty$ as $\tau \rightarrow -\infty$, and $\alpha \rightarrow (t - 2)$ as $\tau \rightarrow t$. Therefore,

$$y(t) = \int_{-\infty}^{t-2} e^{-(t-2-\alpha)} x(\alpha) d\alpha = \int_{-\infty}^{\infty} e^{-(t-2-\alpha)} u(t-2-\alpha) x(\alpha) d\alpha$$

Comparing with

$$y(t) = \int_{-\infty}^{\infty} h(t - \alpha) x(\alpha) d\alpha$$

we get

$$h(t - \alpha) = e^{-(t-2-\alpha)} u(t - 2 - \alpha)$$

Therefore, the impulse response of the system is

$$h(t) = e^{-(t-2)} u(t - 2)$$

Example 2.13 Suppose that the signal $x(t) = u(t + 0.5) - u(t - 0.5)$ and the signal $h(t) = e^{j\omega_0 t}$.

- (a) Determine a value of ω_0 which ensures that $y(0) = 0$, where $y(t) = x(t) * h(t)$.
 (b) Is your answer to the previous part unique?

Solution

Given that

$$x(t) = u(t + 0.5) - u(t - 0.5) = \begin{cases} 1, & -0.5 < t < 0.5 \\ 0, & \text{elsewhere} \end{cases}$$

By definition

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = \int_{-0.5}^{0.5} e^{j\omega_0(t-\tau)} d\tau$$

Therefore,

$$y(t) \Big|_{t=0} = y(0) = \int_{-0.5}^{0.5} e^{-j\omega_0 \tau} d\tau = \frac{2}{\omega_0} \sin\left(\frac{\omega_0}{2}\right)$$

- (a) If $\omega_0 = 2\pi$, then $y(0) = 0$.
 (b) Our answer to part (a) is not unique. For any $\omega_0 = 2n\pi$, $n = \pm 1, \pm 2, \dots$, $y(0) = 0$.

Example 2.14 Consider an LTI system S and a signal $x(t) = 2e^{-3t}u(t - 1)$. If

$$x(t) \longrightarrow y(t) \quad \text{and} \quad \frac{dx(t)}{dt} \longrightarrow -3y(t) + e^{-2t}u(t)$$

then determine the impulse response $h(t)$ of S .

Solution

Given that

$$x(t) = 2e^{-3t}u(t - 1)$$

By differentiating both sides of the above equation w.r.t. t , we get

$$\frac{dx(t)}{dt} = -6e^{-3t}u(t - 1) + 2e^{-3t}\delta(t - 1) = -3x(t) + 2e^{-3t}\delta(t - 1)$$

The given system S is an LTI system. Therefore,

$$x(t) \longrightarrow y(t)$$

$$\delta(t) \longrightarrow h(t)$$

$$2\delta(t-1) \longrightarrow 2h(t-1)$$

$$\frac{dx(t)}{dt} = -3x(t) + 2e^{-3}\delta(t-1) \longrightarrow -3y(t) + 2e^{-3}h(t-1)$$

From the given information, we have

$$-3y(t) + 2e^{-3}h(t-1) = -3y(t) + e^{-2t}u(t)$$

$$2e^{-3}h(t-1) = e^{-2t}u(t)$$

$$h(t-1) = \frac{1}{2}e^3e^{-2t}u(t)$$

$$h(t) = \frac{1}{2}e^3e^{-2(t+1)}u(t+1)$$

$$= \frac{1}{2}e^{-2t}u(t+1)$$

$$h(t) = \frac{1}{2}e^{-2t+1}u(t+1)$$

2.4 DISCRETE-TIME LTI SYSTEMS: THE CONVOLUTION SUM

2.4.1 Convolution Sum

In this section we follow a procedure parallel to that used in the continuous-time case by expressing an arbitrary input $x(n)$ as a sum of impulse components. In Eq. (1.26) we demonstrated the product of $x(n)$ and a time-shifted impulse sequence, i.e., $x(n)\delta(n-k) = x(k)\delta(n-k)$. The component of $x(n)$ at $n=k$ is $x(k)\delta(n-k)$. We see that the multiplication of a signal by a time-shifted impulse results in a time-shifted impulse with amplitude given by the value of the signal at that time the impulse occurs. This property allows us to express $x(n)$ as the following weighted sum of time-shifted impulses:

$$\begin{aligned} x(n) = & \cdots + x(-2)\delta(n+2) + x(-1)\delta(n+1) + x(0)\delta(n) + x(1)\delta(n-1) \\ & + x(2)\delta(n-2) + \cdots \end{aligned}$$

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k) \quad (2.11)$$

A graphical representation of Eq. (2.11) is given in Fig. 2.31.

For a linear system, knowing the system response to impulse $\delta(n)$, the system response to any arbitrary input could be obtained by summing the system response to various

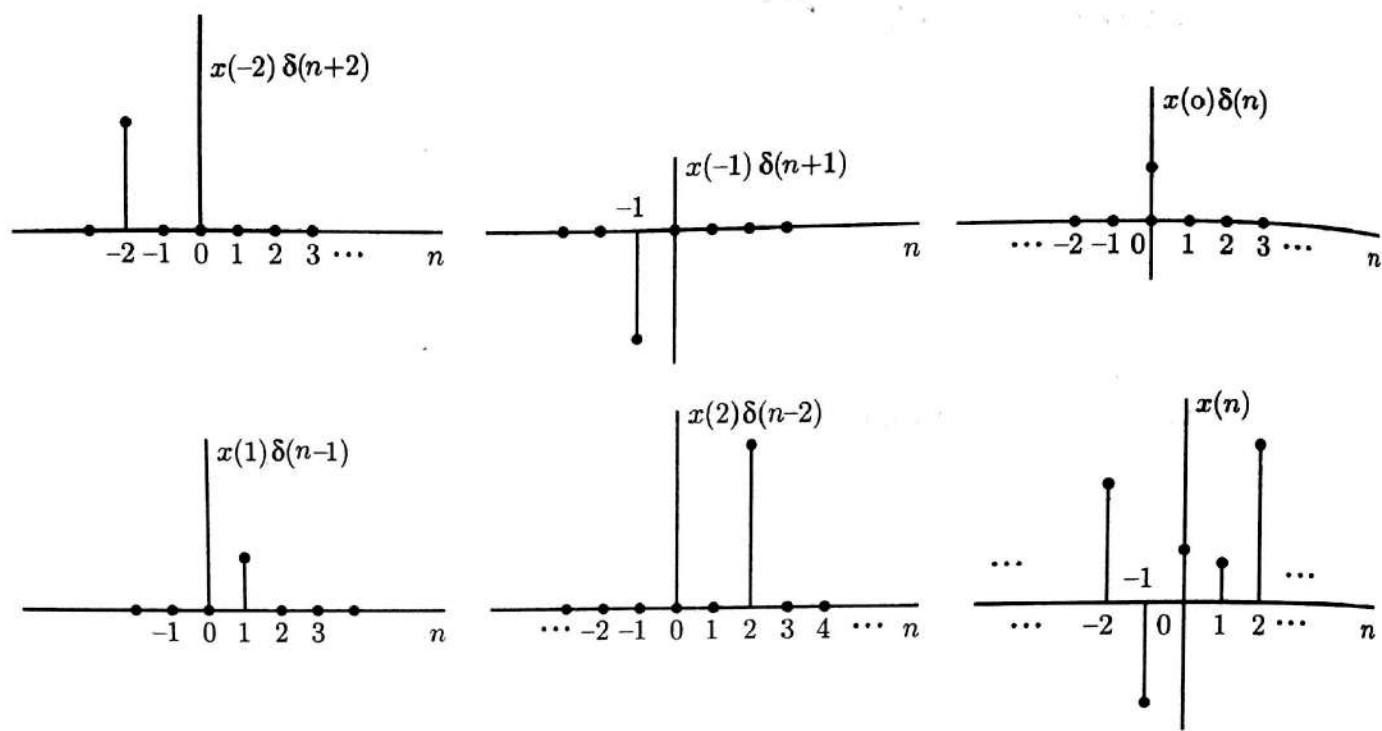


Fig. 2.31 Decomposition of a discrete-time signal into a weighted sum of shifted impulses

impulse components. Let $h(n)$ be the system response to impulse input $\delta(n)$. We shall use the notation

$$x(n) \rightarrow y(n)$$

to indicate the input and the corresponding response of the system. Using the superposition and time-invariant property of LTI systems, we can express output $y(t)$ as a linear combination of the responses of the system to shifted impulse signals.

$$\begin{aligned} \delta(n) &\rightarrow h(n) \\ \delta(n - k) &\rightarrow h(n - k) \\ x(k)\delta(n - k) &\rightarrow x(k)h(n - k) \\ x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n - k) &\rightarrow y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n - k) \end{aligned}$$

Thus, we have obtained the system output $y(n)$ to an arbitrary input $x(n)$ in terms of the unit impulse response $h(n)$:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n - k) \tag{2.12}$$

The relationship expressed in Eq. (2.12) is called the convolution sum of sequences $x(n)$ and $h(n)$. This operation is represented symbolically as

$$y(n) = x(n) * h(n) \tag{2.13}$$

Discrete-convolution (convolution sum) Evaluation

The following examples serve to illustrate techniques that can be used to evaluate discrete convolution. The first example uses two sequences, each defined by a single equation for all values of n .

Example 2.15 Find the convolution of the two sequences

$$x(n) = e^{-n^2}, \text{ for all } n \quad \text{and} \quad h(n) = 3n^2, \text{ for all } n$$

Solution

By definition

$$\begin{aligned} y(n) &= x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) \\ &= \sum_{k=-\infty}^{\infty} e^{-k^2} [3(n-k)^2] \\ &= \sum_{k=-\infty}^{\infty} e^{-k^2} [3(n^2 - 2nk + k^2)] \\ y(n) &= 3n^2 \sum_{k=-\infty}^{\infty} e^{-k^2} - 6n \sum_{k=-\infty}^{\infty} k e^{-k^2} + 3 \sum_{k=-\infty}^{\infty} k^2 e^{-k^2} \\ &= 3n^2 \sum_{k=-\infty}^{\infty} e^{-k^2} - 0 + 3 \sum_{k=-\infty}^{\infty} k^2 e^{-k^2} \\ &= 5.318n^2 + 2.654, \quad \text{for all } n \end{aligned}$$

Example 2.16 Find the convolution of the two sequences

$$x(n) = e^{-n^2}, \text{ for all } n \quad \text{and} \quad h(n) = \begin{cases} e^n, & n < 0 \\ e^{-n}, & n \geq 0 \end{cases}$$

Solution

By definition

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

From the given $x(n)$ and $h(n)$, we get

$$x(k) = e^{-k^2}, \text{ for all } k \quad \text{and} \quad h(n-k) = \begin{cases} e^{n-k}, & (n-k) < 0 \longrightarrow k > n \\ e^{-(n-k)}, & (n-k) \geq 0 \longrightarrow k \leq n \end{cases}$$

Therefore,

$$y(n) = \sum_{k=-\infty}^n e^{-k^2} e^{-(n-k)} + \sum_{k=n+1}^{\infty} e^{-k^2} e^{(n-k)} = e^{-n} \sum_{k=-\infty}^n e^{(k-k^2)} + e^n \sum_{k=n+1}^{\infty} e^{-(k+k^2)}$$

In the previous two examples the need to compute infinite sums creates a difficulty in obtaining solutions for discrete convolutions. The infinite sums do not occur if sequences have zero values prior to some starting time. We show that if

$$x(n) = 0, \quad n < k_1 \quad \text{and} \quad h(n) = 0, \quad n < k_2$$

then

$$x(k) = 0, \quad k < k_1 \quad \text{and} \quad h(n-k) = 0, \quad (n-k) < k_2 \longrightarrow k > (n-k_2)$$

Therefore,

$$y(n) = x(n) * h(n) = \begin{cases} \sum_{k=k_1}^{n-k_2} x(k)h(n-k), & n \geq k_1 + k_2 \\ 0, & n < k_1 + k_2 \end{cases}$$

which is a finite sum.

Example 2.17 Find the convolution of the two sequences

$$x(n) = \begin{cases} 0, & n < -5 \\ \left(\frac{1}{2}\right)^n, & n \geq -5 \end{cases} \quad \text{and} \quad h(n) = \begin{cases} 0, & n < 3 \\ \left(\frac{1}{3}\right)^n, & n \geq 3 \end{cases}$$

Solution

By definition

$$y(n) = x(n) * h(n) = \sum_{k=k_1}^{n-k_2} x(k)h(n-k)$$

where $k_1 = -5$ and $k_2 = 3$. Therefore,

$$y(n) = 0, \quad n < (k_1 + k_2) \longrightarrow n < (-5 + 3) = -2$$

and

$$\begin{aligned} y(n) &= \sum_{k=k_1}^{n-k_2} x(k)h(n-k), \quad n \geq k_1 + k_2 \\ &= \sum_{k=-5}^{n-3} \left(\frac{1}{2}\right)^k \left(\frac{1}{3}\right)^{(n-k)}, \quad n \geq (-5 + 3) = -2 \end{aligned}$$

A change of variables is performed by letting $m = k + 5$, which also yields $m \rightarrow 0$ as $k \rightarrow -5$ and $m \rightarrow (n + 2)$ as $k \rightarrow (n - 3)$. Therefore,

$$\begin{aligned} y(n) &= \sum_{m=0}^{n+2} \left(\frac{1}{2}\right)^{(m-5)} \left(\frac{1}{3}\right)^{(n-m+5)} = \left(\frac{1}{3}\right)^n \left(\frac{2}{3}\right)^5 \sum_{m=0}^{n+2} \left(\frac{3}{2}\right)^m = \left(\frac{1}{3}\right)^n \left(\frac{2}{3}\right)^5 \frac{1 - (3/2)^{n+3}}{1 - 3/2} \\ &= -2 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^n \left[1 - \left(\frac{3}{2}\right)^{n+3}\right] = -2 \left(\frac{2}{3}\right)^5 \left[\left(\frac{1}{3}\right)^n - \left(\frac{1}{2}\right)^n \left(\frac{3}{2}\right)^3\right] \\ &= -64 \left(\frac{1}{3}\right)^{n+5} + 2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{2}\right)^n \end{aligned}$$

Graphical Method for the Convolution Sum

The convolution sum of two signals $x(n)$ and $h(n)$ is given by

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

The convolution operation is performed as follows:

1. We first plot $x(k)$ and $h(k)$ as functions of k (not n) because summation is over k .
2. *Shifting* Shift $h(k)$ by n units to obtain $h(k+n)$.
3. *Folding* Invert $h(k+n)$ about the vertical axis ($k=0$) to obtain $h(-k+n) = h(n-k)$.
4. *Multiplication and addition* Next we multiply $x(k)$ and $h(n-k)$ and add all the products to obtain $y(n)$. The procedure is repeated for each value of n over the range $-\infty$ to ∞ .

We shall demonstrate by an example the graphical procedure for finding the convolution sum.

Example 2.18 Find the convolution sum of a rectangle signal (or gate function) with itself.

$$x(n) = \text{rect}\left(\frac{n}{2N}\right) = \Pi\left(\frac{n}{2N}\right) = \begin{cases} 1, & -N \leq n \leq N \\ 0, & \text{otherwise} \end{cases}$$

Or, evaluate the convolution sum for a system with input $x(n)$ and impulse response $h(n)$, respectively, given by

$$x(n) = h(n) = [u(n+N) - u(n-N-1)]$$

Solution

By definition

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

Step 1: Write the equations and draw sketches for $x(n)$ and $h(n)$ as shown in Fig. 2.32.

$$x(n) = \begin{cases} 1, & -N \leq n \leq N \\ 0, & \text{otherwise} \end{cases} \quad h(n) = \begin{cases} 1, & -N \leq n \leq N \\ 0, & \text{otherwise} \end{cases}$$

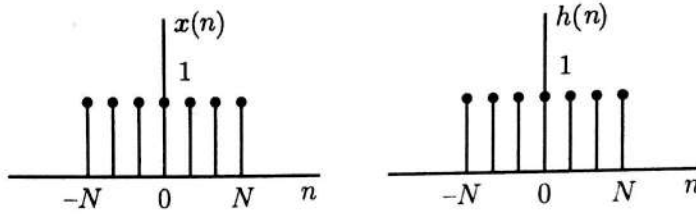


Fig. 2.32 Step 1 for Example 2.18

Step 2: Write the equations and draw sketches for $x(k)$ and $h(n-k)$ [apply precedence rule: $h(n) \rightarrow h(k) \rightarrow h(k+n) \rightarrow h(-k+n) = h(n-k)$] as shown in Fig. 2.33.

$$x(k) = \begin{cases} 1, & -N \leq k \leq N \\ 0, & \text{otherwise} \end{cases} \quad h(n-k) = \begin{cases} 1, & -N \leq (n-k) \leq N \rightarrow (n-N) \leq k \leq (n+N) \\ 0, & \text{otherwise} \end{cases}$$

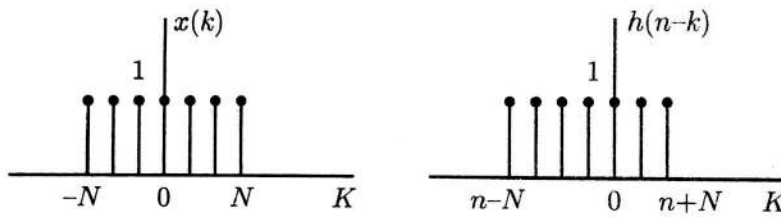


Fig. 2.33 Step 2 for Example 2.18

Step 3: Perform summation to find $y(n)$ for all values of n .

Case I: $(n+N) < -N \rightarrow n < -2N$. Figure 2.34 shows $x(k)$ and $h(n-k)$. We see that the signals do not overlap (Fig. 2.34); hence, the product of the summand is zero.

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) = 0$$

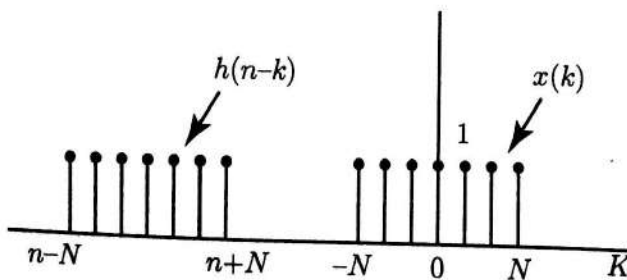


Fig. 2.34 Signal sketch for step 3, case-I

Case II: $-N \leq (n + N) < N \rightarrow -2N \leq n < 0$. Figure 2.35 shows $x(k)$ and $h(n - k)$. The two signals overlap over the interval $-N \leq k \leq (n + N)$ such that

$$y(n) = \sum_{k=-N}^{n+N} 1 = (n + N) - (-N) + 1 = n + 2N + 1, \quad -2N \leq n < 0$$

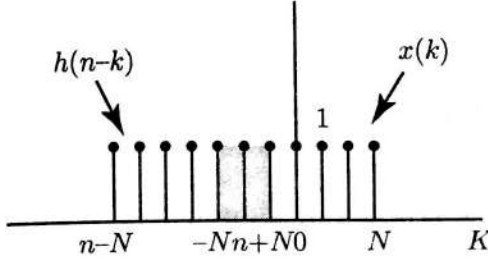


Fig. 2.35 Signal sketch for step 3, case II

Case III: $-N \leq (n - N) \leq N \rightarrow 0 \leq n \leq 2N$. Figure 2.36 shows $x(k)$ and $h(n - k)$. The two signals overlap over the interval $(n - N) \leq k \leq N$, such that

$$y(n) = \sum_{k=n-N}^N 1 = (N) - (n - N) + 1 = 2N - n + 1, \quad 0 \leq n \leq 2N$$

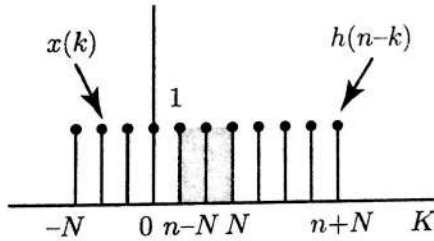


Fig. 2.36 Signal sketch for step 3, case III

Case IV: $(n - N) > N \rightarrow n > 2N$. Figure 2.37 shows $x(k)$ and $h(n - k)$.

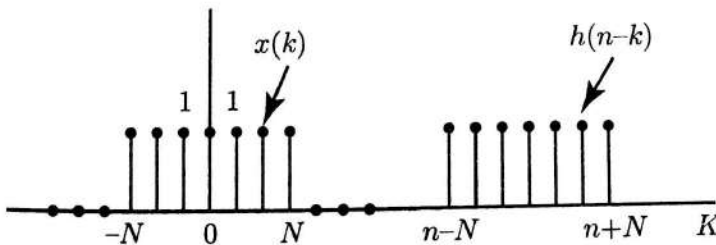


Fig. 2.37 Signal sketch for step 3, case IV

We see that the signals do not overlap (Fig. 2.37); hence, the product of the summand is zero.

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n - k) = 0, \quad n > 2N$$

In summary

$$y(n) = \begin{cases} 0, & n < -2N \\ 2N + n + 1, & -2N \leq n < 0 \\ 2N - n + 1, & 0 \leq n \leq 2N \\ 0, & n > 2N \end{cases}$$

or, in more compact form

$$y(n) = (2N + 1)\text{tri}\left(\frac{n}{2N + 1}\right) = \begin{cases} 2N - |n| + 1, & |n| \leq 2N \\ 0, & |n| \geq 2N \end{cases}$$

This signal (as shown in Fig. 2.38) is called the triangular signal.

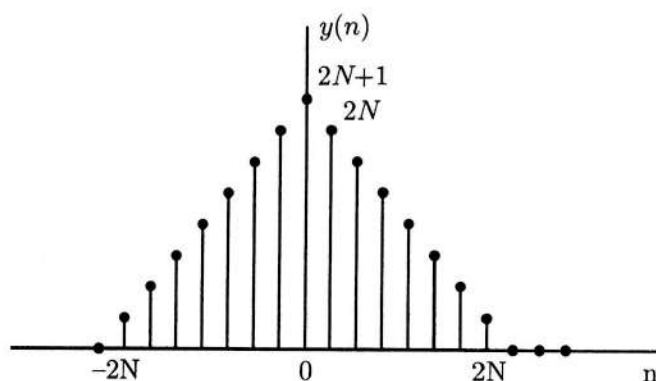


Fig. 2.38 Convolution result

Example 2.19 Compute the convolution $y(n) = x(n) * h(n)$ of the following pairs of signals:

- (a) $x(n) = h(n) = u(n)$
- (b) $x(n) = (0.8)^n u(n)$ and $h(n) = (0.4)^n u(n)$
- (c) $x(n) = h(n) = a^n u(n)$
- (d) $x(n) = u(n - 1)$ and $h(n) = \alpha^n u(n - 1)$
- (e) $x(n) = r(n) = nu(n)$ and $h(n) = a^{-n} u(n - 1)$, where $a < 1$

Solution

By definition

(a)

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) = \sum_{k=-\infty}^{\infty} u(k)u(n-k)$$

The lower limit on the convolution sum simplifies to $k = 0$ [because $u(k) = 0, k < 0$], the upper limit to $k = n$ [because $u(n-k) = 0, k > n$], and we get

$$y(n) = \sum_{k=0}^n 1 = (n+1)u(n) = r(n+1)$$

(b) By definition

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) = \sum_{k=-\infty}^{\infty} (0.8)^k u(k)(0.4)^{n-k} u(n-k)$$

The lower limit on the convolution sum simplifies to $k = 0$ [because $u(k) = 0, k < 0$], the upper limit to $k = n$ [because $u(n-k) = 0, k > n$], and we get

$$\begin{aligned} y(n) &= \sum_{k=0}^n (0.8)^k (0.4)^{n-k} = (0.4)^n \sum_{k=0}^n 2^k = (0.4)^n \frac{1-2^{n+1}}{1-2} = (0.4)^n (2^{n+1} - 1) \\ &= [2(0.8)^n - (0.4)^n] u(n) \end{aligned}$$

(c) By definition

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) = \sum_{k=-\infty}^{\infty} a^k u(k) a^{n-k} u(n-k)$$

The lower limit on the convolution sum simplifies to $k = 0$ [because $u(k) = 0, k < 0$], the upper limit to $k = n$ [because $u(n-k) = 0, k > n$], and we get

$$y(n) = \sum_{k=0}^n a^k a^{n-k} = a^n \sum_{k=0}^n 1 = (n+1)a^n u(n)$$

(d) Given that

$$x(n) = u(n-1) = \begin{cases} 1, & n \geq 1 \\ 0, & n < 1 \end{cases} \quad \text{and} \quad h(n) = \alpha^n u(n-1) = \begin{cases} \alpha^n, & n \geq 1 \\ 0, & n < 1 \end{cases}$$

By definition

$$y(n) = x(n) * h(n) = \sum_{k=k_1}^{n-k_2} x(k)h(n-k)$$

where $k_1 = 1$ and $k_2 = 1$.

$$y(n) = 0 \quad n < (k_1 + k_2) \rightarrow n < 2$$

and

$$\begin{aligned} y(n) &= \sum_{k=k_1}^{n-k_2} x(k)h(n-k) \quad n \geq (k_1 + k_2) \\ &= \sum_{k=1}^{n-1} \alpha^{n-k} \quad n \geq 1 + 1 = 2 \\ &= \alpha^n \sum_{k=1}^{n-1} \alpha^{-k} = \alpha^n \sum_{m=0}^{n-2} \alpha^{-(m+1)} = \alpha^n \alpha^{-1} \sum_{m=0}^{n-2} \alpha^{-(m)} = \frac{\alpha - \alpha^n}{1 - \alpha} \end{aligned}$$

Therefore,

$$y(n) = \begin{cases} \frac{\alpha - \alpha^n}{1 - \alpha}, & n \geq 2 \\ 0, & n < 0 \end{cases} \quad \text{or, more compactly} \quad y(n) = \frac{\alpha - \alpha^n}{1 - \alpha} u(n-2)$$

(e) Given that

$$x(n) = nu(n) = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad \text{and} \quad h(n) = a^{-n}u(n-1) = \begin{cases} a^{-n}, & n \geq 1 \\ 0, & n < 1 \end{cases}$$

By definition

$$y(n) = x(n) * h(n) = \sum_{k=k_1}^{n-k_2} x(k)h(n-k)$$

where $k_1 = 0$ and $k_2 = 1$.

$$y(n) = 0 \quad n < (k_1 + k_2) \rightarrow n < 1$$

and

$$\begin{aligned} y(n) &= \sum_{k=k_1}^{n-k_2} x(k)h(n-k) \quad n \geq (k_1 + k_2) \\ &= \sum_{k=0}^{n-1} ka^{-(n-k)} \quad n \geq (0 + 1) = 1 \\ &= a^{-n} \sum_{k=0}^{n-1} ka^k = \frac{a^{-n+1}}{(1-a)^2} [1 - na^{n-1} + (n-1)a^n] \end{aligned}$$

Therefore,

$$y(n) = \begin{cases} \frac{a^{-n+1}}{(1-a)^2} [1 - na^{n-1} + (n-1)a^n], & n \geq 1 \\ 0, & n < 1 \end{cases}$$

or, more compactly

$$y(n) = \frac{a^{-n+1}}{(1-a)^2} [1 - na^{n-1} + (n-1)a^n] u(n-1)$$

Example 2.20 Suppose that the unit impulse response of an LTI system is a unit ramp,

$$h(n) = r(n) = nu(n)$$

Compute the response of this system to a unit step input $x(n) = u(n)$. Or, determine the step response $s(n)$ of the LTI system characterized by the impulse response $h(n) = nu(n)$.

Solution

Given that

$$x(n) = u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad \text{and} \quad h(n) = nu(n) = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

By definition

$$y(n) = x(n) * h(n) = \sum_{k=k_1}^{n-k_2} x(k)h(n-k)$$

where $k_1 = 0$ and $k_2 = 0$.

$$y(n) = 0 \quad n < (k_1 + k_2) \rightarrow n < 0$$

and

$$\begin{aligned} y(n) &= \sum_{k=k_1}^{n-k_2} x(k)h(n-k), \quad n \geq k_1 + k_2 \\ &= \sum_{k=0}^n (n-k), \quad n \geq 0 \end{aligned}$$

$$y(n) = n + (n-1) + (n-2) + \cdots + 1 + 0 = \frac{n(n+1)}{2}$$

Therefore,

$$y(n) = \begin{cases} \frac{n(n+1)}{2}, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad \text{or, more compactly} \quad y(n) = \frac{n(n+1)}{2} u(n)$$

2.5 PROPERTIES OF THE CONVOLUTION SUM

The structure of the convolution sum is similar to that of the convolution integral. Moreover, the properties of the convolution sum are similar to those of the convolution integral. We shall enumerate these properties here without proofs. The proofs are similar to those for the convolution integral.

2.5.1 Commutative Property

The convolution sum possesses the commutative property:

$$x(n) * h(n) = h(n) * x(n) \quad (2.14)$$

2.5.2 Associative Property

The convolution sum possesses the associative property:

$$x(n) * [h_1(n) * h_2(n)] = [x(n) * h_1(n)] * h_2(n) \quad (2.15)$$

2.5.3 Distributive Property

The convolution sum possesses the distributive property:

$$x(n) * [h_1(n) + h_2(n)] = [x(n) * h_1(n)] + [x(n) * h_2(n)] \quad (2.16)$$

2.5.4 Shifting Property

According to the shift property:

$$\text{if } x(n) * h(n) = y(n), \quad \text{then } x(n) * h(n - n_0) = x(n - n_0) * h(n) = y(n - n_0) \quad (2.17)$$

$$\text{and} \quad x(n - n_1) * h(n - n_2) = y(n - n_1 - n_2) \quad (2.18)$$

2.5.5 Convolution with an Impulse

Convolution of a signal $x(n)$ with a unit impulse function results in the signal $x(n)$ itself:

$$x(n) * \delta(n) = x(n) \quad (2.19)$$

2.5.6 Width Property

The discrete convolution of two sequences has the same type of edge properties as the continuous convolution of two continuous-time signals. That is, the location of the first nonzero sequence value (left edge) of the discrete convolution of two sequences equals the sum of the locations of the first nonzero sequence values (left edges) of the two sequences (i.e., $LE_y = LE_x + LE_h$). The same is true for the last nonzero sequence values (right edges) (i.e., $RE_y = RE_x + RE_h$).

We earlier found that the width of the nonzero extent of the continuous convolution of two signals is the sum of the widths of the nonzero extents of the two signals. A similar property that holds for discrete convolution is that the number of samples in the nonzero extent of a discrete convolution of two sequences is one less than the sum of the number of samples in the nonzero extents of the two sequences. Let N_x and N_h be the number of samples in the nonzero extents of the sequences $x(n)$ and $h(n)$, respectively. The number of samples in the nonzero extents of the sequence $y(n) = x(n) * h(n)$ is

$$N_y = N_x + N_h - 1 \quad (2.20)$$

Proof The number of samples in the nonzero extents of the sequence $x(n)$ is

$$N_x = RE_x - LE_x + 1$$

The number of samples in the nonzero extents of the sequence $h(n)$ is

$$N_h = RE_h - LE_h + 1$$

Since $RE_y = RE_x + RE_h$ and $LE_y = LE_x + LE_h$

The number of samples in the nonzero extents of the sequence $y(n) = x(n) * h(n)$ is

$$\begin{aligned} N_y &= RE_y - LE_y + 1 \\ &= RE_x + RE_h - LE_x - LE_h + 1 \\ &= (RE_x - LE_x + 1) + (RE_h - LE_h + 1) - 1 \\ &= N_x + N_h - 1 \end{aligned}$$

2.5.7 Sum Property

The sum of the impulses in a convolution sum of two discrete-time sequences is the product of the sums of the impulses in the two individual sequences. Let the sum of all the impulses in the signals $y(n)$, $x(n)$, and $h(n)$ be S_y , S_x , and S_h , respectively. Then

$$S_y = S_x S_h \quad (2.21)$$

Proof By definition

$$\begin{aligned} y(n) &= x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) \\ S_y &= \sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x(k)h(n-k) \end{aligned}$$

Interchanging the order of the summation,

$$S_y = \underbrace{\sum_{k=-\infty}^{\infty} x(k)}_{S_x} \underbrace{\sum_{n=-\infty}^{\infty} h(n-k)}_{S_h} = S_x S_h \quad \blacksquare$$

2.6 RELATIONSHIP BETWEEN LTI SYSTEM PROPERTIES AND THE IMPULSE RESPONSE

The impulse response completely characterizes the input-output behaviour of an LTI system. Hence, the properties of the system, such as memory, causality, and stability, are related to the system's impulse response.

2.6.1 LTI Systems With and Without Memory

The present output of a memoryless LTI system depends only on the present input. Exploiting the commutative property of convolution, we may express the output of a discrete-time LTI system as

$$\begin{aligned} y(n) &= h(n) * x(n) \\ &= \sum_{k=-\infty}^{\infty} h(k)x(n-k) \end{aligned}$$

$$= \cdots + h(-2)x(n+2) + h(-1)x(n+1) + h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \cdots$$

For this system to be memoryless, $y(n)$ must depend only on $x(n)$ and therefore cannot depend on $x(n-k)$ for $k \neq 0$. Hence, every term in the above equation must be zero, except $h(0)x(n)$. This condition implies that $h(k) = 0$ for $k \neq 0$; thus a discrete-time LTI system is memoryless if and only if

$$h(n) = K\delta(n) \quad (2.22)$$

where $K = h(0)$ is a constant and the convolution sum reduces to the relation

$$y(n) = Kx(n)$$

If a discrete-time LTI system has an impulse response $h(n)$ that is not identically zero for $n \neq 0$, then the system has memory.

From Eq. (2.1), we can deduce similar properties for continuous-time LTI systems with and without memory. In particular, a continuous-time LTI system is memoryless if $h(t) = 0$ for $t \neq 0$, and such a memoryless LTI system has the form

$$y(t) = Kx(t)$$

for some constant K and has the impulse response

$$h(t) = K\delta(t) \quad (2.23)$$

2.6.2 Causality for LTI Systems

The output of a causal system depends only on the present and past values of the input to the system. Again, we write the convolution sum as

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} h(k)x(n-k) = \left[\sum_{k=-\infty}^{-1} h(k)x(n-k) \right] + \left[\sum_{k=0}^{\infty} h(k)x(n-k) \right] \\ y(n) &= [\dots + h(-2)x(n+2) + h(-1)x(n+1)] + [h(0)x(n) + h(1)x(n-1) \\ &\quad + h(2)x(n-2) + \dots] \end{aligned}$$

We see that past and present values of the input $x(n), x(n-1), x(n-2), \dots$ are associated with indices $k \geq 0$ in the impulse response $h(k)$, while future values of the input, $x(n+1), x(n+2), \dots$ are associated with indices $k < 0$. In order, then, for $y(n)$ to depend only on past and present values of the input, we require that $h(k) = 0$ for $k < 0$. Hence, for a discrete-time causal LTI system,

$$h(n) = 0, \quad \text{for } n < 0 \quad (2.24)$$

and the discrete convolution takes the new form

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k)$$

The alternative equivalent form is

$$y(n) = \sum_{k=-\infty}^n x(k)h(n-k)$$

Similarly, a continuous-time LTI system is causal if

$$h(t) = 0, \quad \text{for } t < 0 \quad (2.25)$$

and in this case the continuous convolution is given by

$$y(t) = \int_0^{\infty} h(\tau)x(t-\tau) d\tau = \int_{-\infty}^t x(\tau)h(t-\tau) d\tau$$

2.6.3 Stability for LTI Systems

A system is stable if every bounded input produces a bounded output. We shall now derive conditions on $h(n)$ that guarantee stability of the system by bounding the convolution sum. The magnitude of the output is given by

$$\begin{aligned} |y(n)| &= |h(n) * x(n)| \\ &= \left| \sum_{k=-\infty}^{\infty} h(k)x(n-k) \right| \\ &\leq \sum_{k=-\infty}^{\infty} |h(k)||x(n-k)| \end{aligned}$$

If we assume that the input is bounded, or $|x(n)| \leq B_x < \infty$, then $|x(n-k)| \leq B_x$, and it follows that

$$|y(n)| \leq B_x \sum_{k=-\infty}^{\infty} |h(k)|$$

From the above equation, we can conclude that if the impulse response is *absolutely summable*, i.e., if

$$\sum_{k=-\infty}^{\infty} |h(k)| < \infty \quad (2.26)$$

then $y(n)$ is bounded in magnitude, and hence, the system is stable.

In continuous time, we obtain an analogous characterization of stability in terms of the impulse response of an LTI system. Specifically, if $|x(t)| \leq B_x$ for all t , then

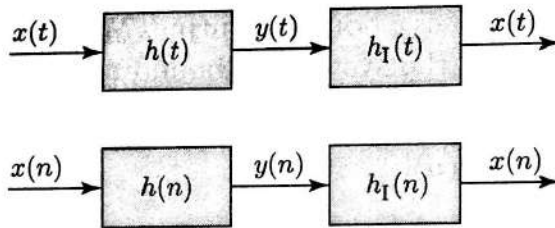
$$\begin{aligned} |y(t)| &= \left| \int_{-\infty}^{\infty} h(\tau)x(t-\tau) d\tau \right| \\ &\leq \int_{-\infty}^{\infty} |h(\tau)||x(t-\tau)| d\tau \\ &\leq B_x \int_{-\infty}^{\infty} |h(\tau)| d\tau \end{aligned}$$

Therefore, the system is stable if the impulse response is *absolutely integrable*, i.e., if

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty \quad (2.27)$$

2.6.4 Invertibility for LTI Systems

Consider a continuous-time LTI system with impulse response $h(t)$. This system is invertible only if an inverse system exists that, when connected in series with the original system, produces an output equal to the input to the first system. Furthermore, if an LTI system is invertible, then it has an LTI inverse.



www.dahiyastech.blogspot.com

Fig. 2.39 Concept of a general invertible system

The relationship between the impulse response, $h(t)$, and that of the corresponding inverse system, $h_I(t)$, is easily derived. The impulse response of the cascade connection in Fig. 2.39 is the convolution of $h(t)$ and $h_I(t)$. We require the output of the cascade to equal the input, or

$$x(t) * [h(t) * h_I(t)] = x(t)$$

This requirement implies that

$$h(t) * h_I(t) = \delta(t) \quad (2.28)$$

similarly, the impulse response of a discrete-time LTI inverse system, $h_I(n)$, must satisfy

$$h(n) * h_I(n) = \delta(n) \quad (2.29)$$

Table 2.1 summarizes the relationship between LTI system properties and impulse response characteristics.

Table 2.1 Properties of the Impulse Response Representation for LTI System

Property	Continuous-time System	Discrete-time System
Memoryless	$h(t) = K\delta(t)$	$h(n) = K\delta(n)$
Causality	$h(t) = 0$, for $t < 0$	$h(n) = 0$, for $n < 0$
Stability	$\int_{-\infty}^{\infty} h(t) dt < \infty$	$\sum_{n=-\infty}^{\infty} h(n) < \infty$
Invertibility	$h(t) * h_I(t) = \delta(t)$	$h(n) * h_I(n) = \delta(n)$

2.6.5 Unit Step Response of an LTI System

Step input signals are often used to characterize the response of an LTI system to sudden changes in the input. The step response $s(n)$ or $s(t)$ is defined as the output due to a unit step input signal $x(n) = u(n)$ or $x(t) = u(t)$. From the convolution sum representation,

the step response of a discrete-time LTI system is the convolution of the unit step with the impulse response, i.e.,

$$s(n) = h(n) * u(n) = \sum_{k=-\infty}^{\infty} h(k)u(n-k)$$

Now, since

$$u(n-k) = \begin{cases} 1, & k \leq n \\ 0, & k > n \end{cases}$$

we have

$$s(n) = \sum_{k=-\infty}^n h(k) \quad (2.30)$$

From this equation, it is clear that $h(n)$ can be recovered from $s(n)$:

$$\begin{aligned} s(n) &= h(n) + \sum_{k=-\infty}^{n-1} h(k) \\ s(n) &= h(n) + s(n-1) \\ h(n) &= s(n) - s(n-1) \end{aligned} \quad (2.31)$$

The step response of a discrete-time LTI system is the running sum of its impulse response. Conversely, the impulse response of a discrete-time LTI system is the first difference of its step response. Similarly, in continuous time, the step response of an LTI system with impulse response $h(t)$ is given by

$$\begin{aligned} s(t) &= \int_{-\infty}^{\infty} h(\tau)u(t-\tau) d\tau \\ s(t) &= \int_{-\infty}^t h(\tau) d\tau \end{aligned} \quad (2.32)$$

i.e., the unit step response of a continuous-time LTI system is the running integral of its impulse response, and from Eq. (2.32), the unit impulse response is the first derivative of the unit step response or

$$h(t) = \frac{ds(t)}{dt} \quad (2.33)$$

Example 2.21 (a) Demonstrate the validity of the following relationships:

$$y(t) = \left(\int_{-\infty}^t x(\tau) d\tau \right) * h'(t) = \int_{-\infty}^t [x'(\tau) * h(\tau)] d\tau = x'(t) * \left(\int_{-\infty}^t h(\tau) d\tau \right)$$

(b) Let $s(t)$ be the unit step response of a continuous-time LTI system. Show that the response $y(t)$ to the input $x(t)$ is

$$y(t) = x'(t) * s(t) = \int_{-\infty}^{\infty} x'(\tau) s(t - \tau) d\tau$$

Solution

(a) We know that

$$y(t) = x(t) * h(t)$$

$$y(t) * u(t) = [x(t) * u(t)] * h(t) \quad (2.34)$$

$$\int_{-\infty}^t y(\tau) d\tau = \left[\int_{-\infty}^t x(\tau) d\tau \right] * h(t)$$

Using the differentiation property of the convolution

$$\begin{aligned} \frac{d}{dt} \left(\int_{-\infty}^t y(\tau) d\tau \right) &= \left[\int_{-\infty}^t x(\tau) d\tau \right] * \frac{d}{dt} h(t) \\ y(t) &= \left[\int_{-\infty}^t x(\tau) d\tau \right] * h'(t) \end{aligned}$$

Now, from Eq. (2.34), we have

$$\begin{aligned} y(t) * u(t) &= x(t) * [h(t) * u(t)] \\ \int_{-\infty}^t y(\tau) d\tau &= x(t) * \left[\int_{-\infty}^t h(\tau) d\tau \right] \end{aligned}$$

Using the differentiation property of the convolution

$$\frac{d}{dt} \left(\int_{-\infty}^t y(\tau) d\tau \right) = \frac{d}{dt} x(t) * \left[\int_{-\infty}^t h(\tau) d\tau \right]$$

$$y(t) = x'(t) * \left[\int_{-\infty}^t h(\tau) d\tau \right] \quad (2.35)$$

Now, by definition,

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} x'(\alpha) \left[\int_{-\infty}^{t-\alpha} h(\tau) d\tau \right] d\alpha = \int_{-\infty}^{\infty} x'(\alpha) \left[\int_{-\infty}^t h(\tau - \alpha) d\tau \right] d\alpha \\ &= \int_{-\infty}^t \left[\int_{-\infty}^{\infty} x'(\alpha) h(\tau - \alpha) d\alpha \right] d\tau = \int_{-\infty}^t [x'(\tau) * h(\tau)] d\tau \end{aligned}$$

(b) Using Eqs (2.32) and (2.35), we have

$$y(t) = x'(t) * s(t) = \int_{-\infty}^{\infty} x'(\tau) s(t - \tau) d\tau$$

www.dahiyastech.blogspot.com

2.7 CORRELATION OF SIGNALS

A mathematical operation that closely resembles convolution is correlation. Just as in the case of convolution two signals are involved in correlation as well. In contrast to convolution, however, our objective in computing the correlation between the two signals is to measure the degree to which the two signals are similar.

2.7.1 Crosscorrelation Function of Energy Signals

Suppose that we have two signals $x(t)$ and $y(t)$ each of which has finite energy. The crosscorrelation of $x(t)$ and $y(t)$ is a signal $R_{xy}(\tau)$, which is defined as

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t) y^*(t - \tau) dt = \int_{-\infty}^{\infty} x(t + \tau) y^*(t) dt \quad (2.36)$$

or, if both signals $x(t)$ and $y(t)$ are real, then

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t) y(t - \tau) dt = \int_{-\infty}^{\infty} x(t + \tau) y(t) dt \quad (2.37)$$

The index τ is the (time) shift (or lag) parameter and the subscripts xy on the crosscorrelation function $R_{xy}(\tau)$ indicate the signals being correlated. The order of the subscripts, with x preceding y , indicates the direction in which one signal is shifted relative to the other.

If we reverse the roles of $x(t)$ and $y(t)$ in Eq. (2.37) and therefore reverse the order of the indices xy , we obtain the crosscorrelation function

$$R_{yx}(\tau) = \int_{-\infty}^{\infty} y(t) x(t - \tau) dt = \int_{-\infty}^{\infty} y(t + \tau) x(t) dt \quad (2.38)$$

By comparing Eq. (2.37) with Eq. (2.38), we conclude that

$$R_{xy}(\tau) = R_{yx}(-\tau) \quad (2.39)$$

Therefore, $R_{xy}(\tau)$ is simply the folded version of $R_{yx}(\tau)$, where the folding is done w.r.t. $\tau = 0$. Hence, $R_{xy}(\tau)$ provides the same information as $R_{yx}(\tau)$ w.r.t. the similarity of $x(t)$ to $y(t)$.

The similarities between the computation of the crosscorrelation of two signals and the convolution integral of two signals is apparent. For energy signals, there is a simple mathematical relationship between correlation and convolution integral:

$$R_{xy}(\tau) = x(\tau) * y(-\tau) \quad (2.40)$$

Proof

$$R_{xy}(\tau) = x(\tau) * y(-\tau) = \int_{-\infty}^{\infty} x(t)y(-(\tau - t)) dt = \int_{-\infty}^{\infty} x(t)y(t - \tau) dt \quad \blacksquare$$

2.7.2 Crosscorrelation Function of Power Signals

The crosscorrelation function between two power signals $x(t)$ and $y(t)$ is mathematically defined by

$$R_{xy}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)y^*(t - \tau) dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t + \tau)y^*(t) dt \quad (2.41)$$

If $x(t)$ and $y(t)$ are both real, then

$$R_{xy}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)y(t - \tau) dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t + \tau)y(t) dt \quad (2.42)$$

An important special case of correlation of power signals is the correlation between two periodic signals whose fundamental periods are such that the product of the two signals is also periodic. This will happen any time the fundamental periods of the two periodic signals have an LCM.

For two signals whose product has a period T , the general form of the correlation function (for real power signals) is given by

$$R_{xy}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)y(t - \tau) dt$$

can be replaced by

$$R_{xy}(\tau) = \frac{1}{T} \int_{-T/2}^{T/2} x(t)y(t - \tau) dt \quad (2.43)$$

because the integral over one period of the product, divided by the period (which is the average of the integrand over one period) is the same as the average over any integer number of periods, including infinitely many periods.

2.7.3 Autocorrelation Function of Continuous-time Signals

A very important special case of the correlation function is the correlation of a function with itself. This type of correlation function is called the *autocorrelation* function. If $x(t)$ is an energy signal, its autocorrelation is

$$R_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)x(t-\tau) dt = \int_{-\infty}^{\infty} x(t+\tau)x(t) dt \quad (2.44)$$

If $x(t)$ is a power signal, its autocorrelation is

$$R_{xx}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t-\tau) dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t+\tau)x(t) dt \quad (2.45)$$

2.7.4 Properties of Crosscorrelation and Autocorrelation Functions

Following are the properties of crosscorrelation and autocorrelation functions:

1. We have already demonstrated that the crosscorrelation function satisfies the property

$$R_{xy}(\tau) = R_{yx}(-\tau)$$

With $y(t) = x(t)$, this relation results in the following important property for the autocorrelation function:

$$R_{xx}(\tau) = R_{xx}(-\tau) \quad (2.46)$$

Hence, the autocorrelation function is an even function.

Proof

$$R_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)x(t-\tau) dt$$

A change of variables is performed by letting $(t-\tau) = \alpha$, which also yields $dt = d\alpha$, $\alpha \rightarrow \infty$ as $t \rightarrow \infty$, and $\alpha \rightarrow -\infty$ as $t \rightarrow -\infty$. Therefore,

$$\begin{aligned} R_{xx}(\tau) &= \int_{-\infty}^{\infty} x(\alpha+\tau)x(\alpha) d\alpha = \int_{-\infty}^{\infty} x(\alpha)x(\alpha+\tau) d\alpha = \int_{-\infty}^{\infty} x(\alpha)x(\alpha-(-\tau)) d\alpha \\ &= R_{xx}(-\tau) \end{aligned}$$

2. Relation to signal energy and signal power:

Case 1 If $x(t)$ is an energy signal, then

$$R_{xx}(0) = E_x = \int_{-\infty}^{\infty} x^2(t) dt \quad (2.47)$$

Proof

$$\begin{aligned} R_{xx}(\tau) &= \int_{-\infty}^{\infty} x(t)x(t-\tau) dt \\ R_{xx}(\tau)|_{\tau=0} &= \int_{-\infty}^{\infty} x(t)x(t) dt \\ R_{xx}(0) &= \int_{-\infty}^{\infty} x^2(t) dt = E_x \end{aligned}$$

Case 2 If $x(t)$ is a power signal, then

$$R_{xx}(0) = P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt \quad (2.48)$$

Proof

$$\begin{aligned} R_{xx}(\tau) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t-\tau) dt \\ R_{xx}(\tau)|_{\tau=0} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t) dt \\ R_{xx}(0) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt = P_x \end{aligned}$$

3. The crosscorrelation satisfies the following condition:

$$|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0)R_{yy}(0)} = \sqrt{E_x E_y} \quad (2.49)$$

The autocorrelation satisfies the following condition:

$$|R_{xx}(\tau)| \leq R_{xx}(0) = E_x \quad (2.50)$$

Proof Let $x(t)$ and $y(t)$ be the two finite energy signals from which we form the linear combination,

$$ax(t) + by(t - \tau)$$

where a and b are two constants and τ is some time shift. The energy in this signal is

$$\begin{aligned} \int_{-\infty}^{\infty} [ax(t) + by(t - \tau)]^2 dt &= a^2 \int_{-\infty}^{\infty} x^2(t) dt + b^2 \int_{-\infty}^{\infty} y^2(t - \tau) dt + 2ab \int_{-\infty}^{\infty} x(t)y(t - \tau) dt \\ &= a^2 R_{xx}(0) + b^2 R_{yy}(0) + 2ab R_{xy}(\tau) \end{aligned}$$

First, we note that $R_{xx}(0) = E_x$ and $R_{yy}(0) = E_y$, which are the energies of $x(t)$ and $y(t)$, respectively. It is obvious that

$$a^2 R_{xx}(0) + b^2 R_{yy}(0) + 2ab R_{xy}(\tau) \geq 0$$

Now, assuming that $b \neq 0$, we can divide the above equation by b^2 to obtain

$$R_{xx}(0) \left(\frac{a}{b}\right)^2 + 2R_{xy}(\tau) \left(\frac{a}{b}\right) + R_{yy}(0) \geq 0$$

We view this equation as a quadratic equation with coefficients $R_{xx}(0)$, $2R_{xy}(\tau)$, and $R_{yy}(0)$. Since the quadratic is nonnegative, it follows that its discriminant must be nonpositive (the discriminant of quadratic polynomial $ax^2 + bx + c$ is $b^2 - 4ac$), i.e.,

$$\begin{aligned} 4[R_{xy}^2(\tau) - R_{xx}(0)R_{yy}(0)] &\leq 0 \\ |R_{xy}(\tau)| &\leq \sqrt{R_{xx}(0)R_{yy}(0)} = \sqrt{E_x E_y} \end{aligned}$$

For autocorrelation function where $y(t) = x(t)$, the above equation reduces to

$$|R_{xx}(\tau)| \leq R_{xx}(0) = E_x$$

This means that the autocorrelation function of a signal attains its maximum value at zero lag. ■

4. The autocorrelation of a periodic signal with period T is periodic with the same period:

$$\text{if } x(t) = x(t \pm T), \quad \text{then } R_{xx}(\tau) = R_{xx}(\tau \pm T) \quad (2.51)$$

Proof Let $x(t)$ be a periodic signal with period T . Then

$$x(t) = x(t + T)$$

$$x(t - \tau) = x(t - \tau + T)$$

Now,

$$\begin{aligned}
 R_{xx}(\tau) &= \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t-\tau) dt \\
 R_{xx}(\tau-T) &= \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t-(\tau-T)) dt \\
 &= \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t-\tau+T) dt = \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t-\tau) dt = R_{xx}(\tau) \quad \blacksquare
 \end{aligned}$$

5. If $R_{xy}(0) = 0$, then $x(t)$ and $y(t)$ are orthogonal.

Proof By definition

$$\begin{aligned}
 R_{xy}(\tau) &= \int_{-\infty}^{\infty} x(t)y(t-\tau) dt \\
 R_{xy}(\tau)|_{\tau=0} &= R_{xy}(0) = \int_{-\infty}^{\infty} x(t)y(t) dt = 0 \quad \blacksquare
 \end{aligned}$$

2.7.5 Crosscorrelation Sequence of Discrete-time Energy Signals

Suppose that we have two signals $x(n)$ and $y(n)$ each of which has finite energy. The crosscorrelation of $x(n)$ and $y(n)$ is a signal $R_{xy}(m)$, which is defined as

$$R_{xy}(m) = \sum_{n=-\infty}^{\infty} x(n)y^*(n-m) = \sum_{n=-\infty}^{\infty} x(n+m)y^*(n) \quad (2.52)$$

or, if both signals $x(n)$ and $y(n)$ are real, then

$$R_{xy}(m) = \sum_{n=-\infty}^{\infty} x(n)y(n-m) = \sum_{n=-\infty}^{\infty} x(n+m)y(n) \quad (2.53)$$

The index m is the (time) shift (or lag) parameter and the subscripts xy on the crosscorrelation function $R_{xy}(m)$ indicate the signals being correlated. The order of the subscripts, with x preceding y , indicates the direction in which one signal is shifted relative to the other.

If we reverse the roles of $x(n)$ and $y(n)$ in Eq. (2.53) and therefore reverse the order of the indices xy , we obtain the crosscorrelation function

$$R_{yx}(m) = \sum_{n=-\infty}^{\infty} y(n)x(n-m) = \sum_{n=-\infty}^{\infty} y(n+m)x(n) \quad (2.54)$$

By comparing Eq. (2.53) with Eq. (2.54), we conclude that

$$R_{xy}(m) = R_{yx}(-m) \quad (2.55)$$

Therefore, $R_{xy}(m)$ is simply the folded version of $R_{yx}(m)$, where the folding is done w.r.t. $m = 0$. Hence, $R_{xy}(m)$ provides the same information as $R_{yx}(m)$ w.r.t. the similarity of $x(n)$ to $y(n)$.

The similarities between the computation of the crosscorrelation of two signals and the convolution sum of two sequences is apparent. For energy signals, there is a simple mathematical relationship between correlation and convolution sum:

$$R_{xy}(m) = x(m) * y(-m) \quad (2.56)$$

Proof

$$R_{xy}(m) = x(m) * y(-m) = \sum_{n=-\infty}^{\infty} x(n)y(-(m-n)) = \sum_{n=-\infty}^{\infty} x(n)y(n-m) \quad \blacksquare$$

2.7.6 Crosscorrelation Sequence of Power Signals

The crosscorrelation function between two power signals $x(n)$ and $y(n)$ is mathematically defined by

$$R_{xy}(m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n)y^*(n-m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n+m)y^*(n) \quad (2.57)$$

If $x(n)$ and $y(n)$ are both real,

$$R_{xy}(m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n)y(n-m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n+m)y(n) \quad (2.58)$$

An important special case of correlation of power signals is the correlation between two periodic signals whose fundamental periods are such that the product of the two signals is also periodic. This will happen any time the fundamental periods of the two periodic signals have an LCM.

For two signals whose product has a period N , the general form of the correlation function (for real power signals)

$$R_{xy}(m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n)y(n-m)$$

can be replaced by

$$R_{xy}(m) = \frac{1}{2N+1} \sum_{n=-N}^N x(n)y(n-m) \quad (2.59)$$

2.7.7 Autocorrelation Sequence of Discrete-time Signals

In the special case where $y(n) = x(n)$, we have the autocorrelation sequence of $x(n)$. If $x(n)$ is an energy signal, its autocorrelation is

$$R_{xx}(m) = \sum_{n=-\infty}^{\infty} x(n)x(n-m) = \sum_{n=-\infty}^{\infty} x(n+m)x(n) \quad (2.60)$$

If $x(n)$ is a power signal, its autocorrelation is

$$R_{xx}(m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n)x(n-m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n+m)x(n) \quad (2.61)$$

2.7.8 Properties of Crosscorrelation and Autocorrelation Sequences

Following are the properties of crosscorrelation and autocorrelation sequences:

1. We have already demonstrated that the crosscorrelation sequence satisfies the property

$$R_{xy}(m) = R_{yx}(-m)$$

With $y(n) = x(n)$, this relation results in the following important property for the autocorrelation sequence:

$$R_{xx}(m) = R_{xx}(-m) \quad (2.62)$$

Hence, The autocorrelation sequence is an even function.

Proof By definition

$$R_{xx}(m) = \sum_{n=-\infty}^{\infty} x(n)x(n-m)$$

A change of variables is performed by letting $n-m = k$, which also yields $k \rightarrow \infty$ as $n \rightarrow \infty$ and $k \rightarrow -\infty$ as $n \rightarrow -\infty$. Therefore,

$$\begin{aligned} R_{xx}(m) &= \sum_{n=-\infty}^{\infty} x(k+m)x(k) = \sum_{n=-\infty}^{\infty} x(k)x(k+m) = \sum_{n=-\infty}^{\infty} x(k)x(k-(-m)) \\ &= R_{xx}(-m) \end{aligned}$$

2. Relation to signal energy and signal power:

Case 1: If $x(n)$ is an energy signal, then

$$R_{xx}(0) = E_x = \sum_{n=-\infty}^{\infty} x^2(n) \quad (2.63)$$

Proof By definition

$$R_{xx}(m) = \sum_{n=-\infty}^{\infty} x(n)x(n-m)$$

$$R_{xx}(m)\Big|_{m=0} = R_{xx}(0) = \sum_{n=-\infty}^{\infty} x(n)x(n) = \sum_{n=-\infty}^{\infty} x^2(n) = E_x \quad \blacksquare$$

Case 2: If $x(n)$ is a power signal, then

$$R_{xx}(0) = P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x^2(n) \quad (2.64)$$

Proof By definition

$$R_{xx}(m) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n)x(n-m)$$

$$R_{xx}(m)\Big|_{m=0} = R_{xx}(0) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n)x(n) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x^2(n) = P_x \quad \blacksquare$$

3. The crosscorrelation sequence satisfies the following condition:

$$|R_{xy}(m)| \leq \sqrt{R_{xx}(0)R_{yy}(0)} = \sqrt{E_x E_y} \quad (2.65)$$

The autocorrelation sequence satisfies the following condition:

$$|R_{xx}(m)| \leq R_{xx}(0) = E_x \quad (2.66)$$

Proof Let $x(n)$ and $y(n)$ be the two finite energy discrete-time signals from which we form the linear combination

$$ax(n) + by(n-m)$$

where a and b are two constants and m is some time shift. The energy in this signal is

$$\begin{aligned} \sum_{n=-\infty}^{\infty} [ax(n) + by(n-m)]^2 &= a^2 \sum_{n=-\infty}^{\infty} x^2(n) + b^2 \sum_{n=-\infty}^{\infty} y^2(n-m) \\ &\quad + 2ab \sum_{n=-\infty}^{\infty} x(n)y(n-m) \\ &= a^2 R_{xx}(0) + b^2 R_{yy}(0) + 2ab R_{xy}(m) \end{aligned}$$

First, we note that $R_{xx}(0) = E_x$ and $R_{yy}(0) = E_y$, which are the energies of $x(n)$ and $y(n)$ respectively. It is obvious that

$$a^2 R_{xx}(0) + b^2 R_{yy}(0) + 2ab R_{xy}(m) \geq 0$$

Now, assuming that $b \neq 0$, we can divide the above equation by b^2 to obtain

$$R_{xx}(0) \left(\frac{a}{b}\right)^2 + 2R_{xy}(m) \left(\frac{a}{b}\right) + R_{yy}(0) \geq 0$$

We view this equation as a quadratic equation with coefficients $R_{xx}(0)$, $2R_{xy}(m)$, and $R_{yy}(0)$. Since the quadratic is nonnegative, it follows that its discriminant must be nonpositive, i.e.,

$$4[R_{xy}^2(m) - R_{xx}(0)R_{yy}(0)] \leq 0$$

$$|R_{xy}(m)| \leq \sqrt{R_{xx}(0)R_{yy}(0)} = \sqrt{E_x E_y}$$

For the autocorrelation function where $y(n) = x(n)$, the above equation reduces to

$$|R_{xx}(m)| \leq R_{xx}(0) = E_x$$

SOLVED EXAMPLES

Example 2.22 Let

$$x(t) = u(t-3) - u(t-5) \text{ and } h(t) = e^{-3t}u(t)$$

- (a) Compute $y(t) = x(t) * h(t)$.
- (b) Compute $g(t) = \left(\frac{d}{dt}x(t)\right) * h(t)$.
- (c) How is $g(t)$ related to $y(t)$?

Solution

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau$$

Step 1: Write the equations and draw sketches for $x(t)$ and $h(t)$ as shown in Fig. 2.40.

$$x(t) = u(t-3) - u(t-5) = \begin{cases} 1, & 3 < t < 5 \\ 0, & \text{otherwise} \end{cases}$$

$$h(t) = \begin{cases} e^{-3t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

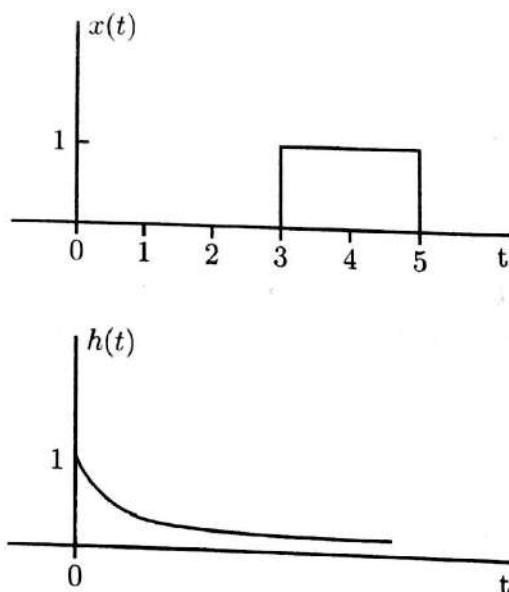


Fig. 2.40 Step 1 for Example 2.22

Step 2: Write the equations and draw sketches for $x(\tau)$ and $h(t-\tau)$ [apply precedence rule: $h(t) \rightarrow h(\tau) \rightarrow h(\tau+t) \rightarrow h(-\tau+t) = h(t-\tau)$] as shown in Fig. 2.41.

$$x(\tau) = \begin{cases} 1, & 3 < \tau < 5 \\ 0, & \text{otherwise} \end{cases}$$

$$h(t-\tau) = \begin{cases} e^{-3(t-\tau)}, & (t-\tau) \geq 0 \rightarrow \tau \leq t \\ 0, & (t-\tau) < 0 \rightarrow \tau > t \end{cases}$$

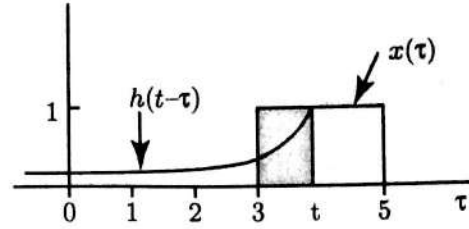


Fig. 2.43 Signal sketch for step 3, case II

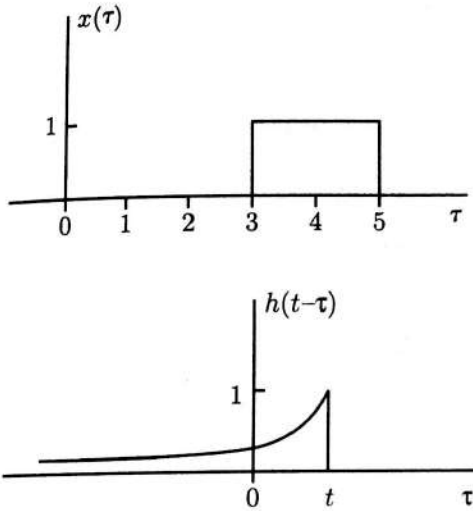


Fig. 2.41 Step 2 for Example 2.22

Step 3: Perform integration to find $y(t)$ for all values of t .

Case I: $t < 3$. Figure 2.42 shows $x(\tau)$ and $h(t-\tau)$. We see that the signals do not overlap (Fig. 2.42); hence, the product of the integrand is zero.

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau = 0$$

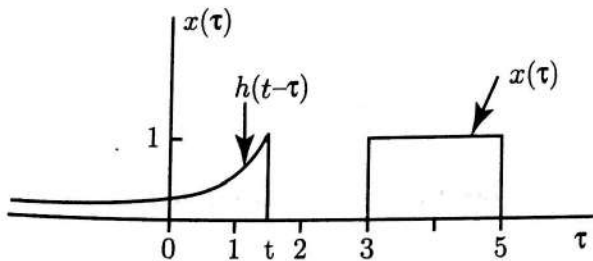


Fig. 2.42 Signal sketch for step 3, case I

Case II: $3 \leq t < 5$. Figure 2.43 shows $x(\tau)$ and $h(t-\tau)$. The two signals overlap over the interval $3 \leq \tau \leq t$ such that

$$y(t) = \int_3^t e^{-3(t-\tau)} d\tau = e^{-3t} \int_3^t e^{3\tau} d\tau$$

$$= \frac{1 - e^{-3(t-3)}}{3}$$

Case III: $t \geq 5$. Figure 2.44 shows $x(\tau)$ and $h(t-\tau)$.

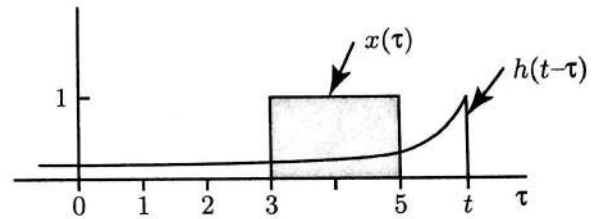


Fig. 2.44 Signal sketch for step 3, case III

The two signals overlap over the interval $3 \leq \tau \leq 5$ such that

$$y(t) = \int_3^5 e^{-3(t-\tau)} d\tau = e^{-3t} \int_3^5 e^{3\tau} d\tau$$

$$= \frac{(1 - e^{-6})e^{-3(t-5)}}{3}$$

In summary

$$y(t) = \begin{cases} 0, & t < 3 \\ \frac{1 - e^{-3(t-3)}}{3}, & 3 \leq t < 5 \\ \frac{(1 - e^{-6})e^{-3(t-5)}}{3}, & t \geq 5 \end{cases}$$

(b) Differentiating $x(t)$ w.r.t. t , we get

$$\frac{dx(t)}{dt} = \delta(t-3) - \delta(t-5)$$

Therefore,

$$\begin{aligned}
 g(t) &= \left(\frac{d}{dt} x(t) \right) * h(t) \\
 &= [\delta(t-3) - \delta(t-5)] * h(t) \\
 &= h(t-3) - h(t-5) \\
 &= e^{-3(t-3)} u(t-3) - e^{-3(t-5)} u(t-5)
 \end{aligned}$$

(c) From the differentiation property, i.e., Eq. (2.9) of the convolution

$$\frac{dy(t)}{dt} = \frac{dx(t)}{dt} * h(t)$$

Hence,

$$g(t) = \frac{dy(t)}{dt}$$

Example 2.23 Let

$$y(t) = e^{-t} u(t) * \sum_{k=-\infty}^{\infty} \delta(t-3k)$$

Show that $y(t) = Ae^{-t}$ for $0 \leq t < 3$, and determine the value of A .

Solution

$$\begin{aligned}
 y(t) &= e^{-t} u(t) * \sum_{k=-\infty}^{\infty} \delta(t-3k) \\
 &= \sum_{k=-\infty}^{\infty} e^{-(t-3k)} u(t-3k) \\
 &= \begin{cases} \sum_{k=-\infty}^{\infty} e^{-t+3k}, & (t-3k) > 0 \longrightarrow t > 3k \\ 0, & \text{otherwise} \end{cases}
 \end{aligned}$$

To determine $y(t)$ over the interval $0 \leq t < 3$, $k = 0, -1, -2, \dots$. Therefore, we have

$$\begin{aligned}
 y(t) &= \sum_{k=-\infty}^0 e^{-(t-3k)}, \quad 0 \leq t < 3 \\
 &= e^{-t} \sum_{k=-\infty}^0 e^{3k} \\
 &= e^{-t} \sum_{k=0}^{\infty} (e^{-3})^k \\
 y(t) &= \frac{e^{-t}}{1 - e^{-3}}, \quad 0 \leq t < 3
 \end{aligned}$$

Therefore, $A = 1/(1 - e^{-3})$.

Example 2.24 The following are the impulse responses of discrete-time LTI systems. Determine whether each system is causal and/or stable. Justify your answers.

- (a) $h(n) = (1/5)^n u(n)$
- (b) $h(n) = (0.8)^n u(n+2)$
- (c) $h(n) = (1/2)^n u(-n)$
- (d) $h(n) = 5^n u(3-n)$
- (e) $h(n) = (-1/2)^n u(n) + (1.01)^n u(n-1)$

Solution

(a) Causal, because $h(n) = 0$ for $n < 0$.

$$\begin{aligned}
 h(n) &= \left(\frac{1}{5} \right)^n u(n) \\
 &= \begin{cases} \left(\frac{1}{5} \right)^n, & n \geq 0 \\ 0, & n < 0 \end{cases}
 \end{aligned}$$

Stable, because

$$\begin{aligned}
 \sum_{n=-\infty}^{\infty} h(n) &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{5} \right)^n u(n) \\
 &= \sum_{n=0}^{\infty} \left(\frac{1}{5} \right)^n = \frac{5}{4} < \infty
 \end{aligned}$$

(b) Noncausal, because $h(n) \neq 0$ for $n < 0$.

$$h(n) = \begin{cases} (0.8)^n, & n \geq -2 \\ 0, & n < -2 \end{cases}$$

Stable, because

$$\begin{aligned}
 \sum_{n=-\infty}^{\infty} h(n) &= \sum_{n=-\infty}^{\infty} (0.8)^n u(n+2) \\
 &= \sum_{n=-2}^{\infty} (0.8)^n = 5 < \infty
 \end{aligned}$$

(c) Anticausal, because $h(n) = 0$ for $n > 0$. Unstable, because

$$\begin{aligned}
 \sum_{n=-\infty}^{\infty} h(n) &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2} \right)^n u(-n) \\
 &= \sum_{n=-\infty}^0 \left(\frac{1}{2} \right)^n = \infty
 \end{aligned}$$

(d) Noncausal, because $h(n) \neq 0$ for $n < 0$.

$$h(n) = \begin{cases} 5^n, & n \leq 3 \\ 0, & n > 3 \end{cases}$$

Stable, because

$$\begin{aligned} \sum_{n=-\infty}^{\infty} h(n) &= \sum_{n=-\infty}^{\infty} 5^n u(3-n) \\ &= \sum_{n=-\infty}^3 5^n = \frac{625}{4} < \infty \end{aligned}$$

(e) Causal, because $h(n) = 0$ for $n < 0$.

$$h(n) = \begin{cases} 1, & n = 0 \\ \left(-\frac{1}{2}\right)^n + (1.01)^n, & n \geq 1 \end{cases}$$

Unstable, because

$$\begin{aligned} \sum_{n=-\infty}^{\infty} h(n) &= \sum_{n=-\infty}^{\infty} \left[\left(-\frac{1}{2}\right)^n u(n) + (1.01)^n u(n-1) \right] \\ &= \sum_{n=-\infty}^{\infty} \left(-\frac{1}{2}\right)^n u(n) + \sum_{n=-\infty}^{\infty} (1.01)^n u(n-1) \\ &= \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n + \sum_{n=1}^{\infty} (1.01)^n = \infty \end{aligned}$$

Example 2.25 The following are the impulse responses of continuous-time LTI systems. Determine whether each system is causal and/or stable. Justify your answers.

(a) $h(t) = e^{-4t}u(t-2)$

(b) $h(t) = e^{-6t}u(3-t)$

(c) $h(t) = e^{-6|t|}$

(d) $h(t) = te^{-t}u(t)$

Solution

(a) Causal, because $h(t) = 0$ for $t < 0$. Stable, because

$$\begin{aligned} \int_{-\infty}^{\infty} |h(t)| dt &= \int_{-\infty}^{\infty} e^{-4t} u(t-2) dt \\ &= \int_2^{\infty} e^{-4t} dt = \frac{e^{-8}}{4} < \infty \end{aligned}$$

(b) Noncausal, because $h(t) \neq 0$ for $t < 0$. Unstable, because $\int_{-\infty}^{\infty} |h(t)| dt = \infty$

(c) Noncausal, because $h(t) \neq 0$ for $t < 0$.

$$\begin{aligned} h(t) &= e^{-6|t|} \\ &= \begin{cases} e^{-6t}, & t > 0 \\ e^{6t}, & t < 0 \end{cases} \\ &= e^{-6t}u(t) + e^{6t}u(-t) \end{aligned}$$

Stable, because

$$\begin{aligned} \int_{-\infty}^{\infty} |h(t)| dt &= \int_{-\infty}^{\infty} e^{-6|t|} dt \\ &= \int_{-\infty}^0 e^{6t} dt + \int_0^{\infty} e^{-6t} dt = \frac{1}{3} < \infty \end{aligned}$$

(d) Causal, because $h(t) = 0$ for $t < 0$. Stable, because

$$\begin{aligned} \int_{-\infty}^{\infty} |h(t)| dt &= \int_{-\infty}^{\infty} te^{-t}u(t) dt \\ &= \int_0^{\infty} te^{-t} dt = 1 < \infty \end{aligned}$$

Example 2.26 Evaluate the step response for the LTI systems represented by the following impulse responses:

(a) $h(n) = (-1/2)^n u(n)$

(b) $h(n) = \delta(n) - \delta(n-2)$

(c) $h(n) = (-1)^n [u(n+2) - u(n-3)]$

(d) $h(t) = e^{-|t|}$

(e) $h(t) = 1/4(u(t) - u(t-4))$

Solution

(a) From Eq. (2.30), we have

$$\begin{aligned} s(n) &= \sum_{k=-\infty}^n h(k) = \sum_{k=-\infty}^n \left(-\frac{1}{2}\right)^k u(k) \\ &= \sum_{k=0}^n \left(-\frac{1}{2}\right)^k = \frac{1 - (-1/2)^{n+1}}{1 - (-1/2)} \\ &= \frac{2}{3} + \frac{1}{3} \left(-\frac{1}{2}\right)^n \end{aligned}$$

(b) From Eq. (2.30), we have

$$\begin{aligned}
 s(n) &= \sum_{k=-\infty}^n h(k) \\
 &= \sum_{k=-\infty}^n [\delta(k) - \delta(k-2)] \\
 &= \sum_{k=-\infty}^n \delta(k) - \sum_{k=-\infty}^n \delta(k-2) \\
 &= u(n) - u(n-2)
 \end{aligned}$$

(c) Again from Eq. (2.30), we have www.dahiyastech.blogspot.com

$$\begin{aligned}
 s(n) &= \sum_{k=-\infty}^n h(k) \\
 &= \sum_{k=-\infty}^n (-1)^k [u(k+2) - u(k-3)] \\
 &= \sum_{k=-\infty}^n (-1)^k u(k+2) - \sum_{k=-\infty}^n (-1)^k u(k-3) \\
 &= \sum_{k=-2}^n (-1)^k - \sum_{k=3}^n (-1)^k \\
 &= \sum_{m=0}^{n+2} (-1)^{m-2} - \sum_{m=0}^{n-3} (-1)^{m+3} \\
 &= \frac{1}{2}[1 + (-1)^n] + \frac{1}{2}[1 - (-1)^n] = 1
 \end{aligned}$$

(d) From Eq. (2.32), we have

$$\begin{aligned}
 s(t) &= \int_{-\infty}^t h(\tau) d\tau = \int_{-\infty}^t e^{-|\tau|} d\tau \\
 &= \int_{-\infty}^0 e^{\tau} d\tau + \int_0^t e^{-\tau} d\tau \\
 &= 2 - e^{-t}
 \end{aligned}$$

(e) From Eq. (2.32), we have

$$\begin{aligned}
 s(t) &= \int_{-\infty}^t h(\tau) d\tau \\
 &= \int_{-\infty}^t \frac{1}{4}(u(\tau) - u(\tau-4)) d\tau \\
 &= \frac{1}{4} \left[\int_{-\infty}^t u(\tau) d\tau - \int_{-\infty}^t u(\tau-4) d\tau \right] \\
 &= \frac{1}{4}[r(t) - r(t-4)]
 \end{aligned}$$

Example 2.27 Consider the signal

$$x(n) = \alpha^n u(n)$$

(a) Determine the signal $g(n) = x(n) - \alpha x(n-1)$.
 (b) Use the result of part (a) in conjunction with properties of convolution in order to determine a sequence $h(n)$ such that

$$x(n) * h(n) = \left(\frac{1}{2}\right)^n [u(n+2) - u(n-2)]$$

Solution

(a) We may write

$$\begin{aligned}
 g(n) &= x(n) - \alpha x(n-1) \\
 &= \alpha^n u(n) - \alpha \alpha^{n-1} u(n-1) \\
 &= \alpha^n u(n) - \alpha^n u(n-1) \\
 &= \alpha^n [u(n) - u(n-1)] \\
 &= \alpha^n \delta(n) = \delta(n)
 \end{aligned}$$

(b) Given that

$$\begin{aligned}
 x(n) * h(n) &= \left(\frac{1}{2}\right)^n [u(n+2) - u(n-2)] \\
 &= \begin{cases} \left(\frac{1}{2}\right)^n, & -2 \leq n \leq 1 \\ 0, & \text{otherwise} \end{cases}
 \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{1}{2}\right)^n [\delta(n+2) + \delta(n+1) \\
&\quad + \delta(n) + \delta(n-1)] \\
&= \left(\frac{1}{2}\right)^n \delta(n+2) + \left(\frac{1}{2}\right)^n \delta(n+1) \\
&\quad + \left(\frac{1}{2}\right)^n \delta(n) + \left(\frac{1}{2}\right)^n \delta(n-1)] \\
&= 4\delta(n+2) + 2\delta(n+1) + \delta(n) + \frac{1}{2}\delta(n-1)
\end{aligned}$$

Note that $g(n) = x(n) * [\delta(n) - \alpha\delta(n-1)]$. Therefore from part (a), we know that

$$x(n) * [\delta(n) - \alpha\delta(n-1)] = \delta(n)$$

Using this and the shift property of the convolution, we may write

$$x(n) * [\delta(n-1) - \alpha\delta(n-2)] = \delta(n-1)$$

$$x(n) * [\delta(n+1) - \alpha\delta(n)] = \delta(n+1)$$

$$x(n) * [\delta(n+2) - \alpha\delta(n+1)] = \delta(n+2)$$

Therefore,

$$\begin{aligned}
&x(n) * h(n) \\
&= 4x(n) * [\delta(n+2) - \alpha\delta(n+1)] \\
&\quad + 2x(n) * [\delta(n+1) - \alpha\delta(n)] \\
&\quad + x(n) * [\delta(n) - \alpha\delta(n-1)] \\
&\quad + \frac{1}{2}x(n) * [\delta(n) - \alpha\delta(n-1)] \\
&= x(n) * [4\delta(n+2) - 4\alpha\delta(n+1) \\
&\quad + 2\delta(n+1) - 2\alpha\delta(n) + \delta(n) \\
&\quad - \alpha\delta(n-1) + \frac{1}{2}\delta(n-1) - \frac{1}{2}\alpha\delta(n-2)] \\
&= x(n) * [4\delta(n+2) + (2-4\alpha)\delta(n+1) \\
&\quad + (1-2\alpha)\delta(n) + (1/2-\alpha)\delta(n-1) \\
&\quad + (1/2)\delta(n-2)]
\end{aligned}$$

Hence,

$$\begin{aligned}
h(n) &= 4\delta(n+2) + (2-4\alpha)\delta(n+1) \\
&\quad + (1-2\alpha)\delta(n) + (1/2-\alpha)\delta(n-1) \\
&\quad + (1/2)\delta(n-2)
\end{aligned}$$

Example 2.28 Consider the cascade of the following two systems S_1 and S_2 as depicted in Fig. 2.45:

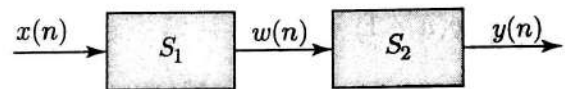


Fig. 2.45

S_1 : causal LTI

$$w(n) = \frac{1}{2}w(n-1) + x(n)$$

S_2 : causal LTI

$$y(n) = \alpha y(n-1) + \beta w(n)$$

The difference equation relating $x(n]$ and $y(n]$ is

$$y(n) = -\frac{1}{8}y(n-2) + \frac{3}{4}y(n-1) + x(n)$$

(a) Determine α and β .

(b) Show the impulse response of the cascade connection of S_1 and S_2 .

Solution

Consider the difference equation relating $y(n]$ and $w(n]$ for S_2 :

$$y(n) = \alpha y(n-1) + \beta w(n)$$

From this equation we may write

$$w(n) = \frac{1}{\beta}y(n) - \frac{\alpha}{\beta}y(n-1)$$

and

$$w(n-1) = \frac{1}{\beta}y(n-1) - \frac{\alpha}{\beta}y(n-2)$$

Substituting this in the difference equation relating $w(n)$ and $x(n)$ for S_1 :

$$\begin{aligned} w(n) &= \frac{1}{2} \left(\frac{1}{\beta} y(n-1) - \frac{\alpha}{\beta} y(n-2) \right) + x(n) \\ &= \frac{1}{2\beta} y(n-1) - \frac{\alpha}{2\beta} y(n-2) + x(n) \end{aligned}$$

Substituting this in the difference equation relating $y(n)$ and $w(n)$ for S_2 :

$$\begin{aligned} y(n) &= \alpha y(n-1) \\ &\quad + \beta \left(\frac{1}{2\beta} y(n-1) - \frac{\alpha}{2\beta} y(n-2) + x(n) \right) \\ y(n) &= \left(\alpha + \frac{1}{2} \right) y(n-1) - \frac{\alpha}{2} y(n-2) + \beta x(n) \end{aligned}$$

Comparing with the given equation relating $y(n)$ and $x(n)$, we obtain

$$\alpha = \frac{1}{4} \quad \text{and} \quad \beta = 1$$

Example 2.29 Determine whether each of the following statements concerning LTI systems is true or false. Justify your answers.

- (a) If $h(t)$ is the impulse response of an LTI system and $h(t)$ is periodic and nonzero, the system is unstable.
- (b) The inverse of a causal LTI system is always causal.
- (c) If $|h(n)| \leq K$ for each n , where K is a given number, then the LTI system with $h(n)$ as its impulse response is stable.
- (d) If a discrete-time LTI system has an impulse response $h(n)$ of finite duration, the system is stable.
- (e) If an LTI system is causal, it is stable.
- (f) The cascade of a noncausal LTI system with a causal one is necessarily noncausal.
- (g) A continuous-time LTI system is stable if and only if its step response $s(t)$ is absolutely integrable, i.e., if and only if

$$\int_{-\infty}^{\infty} |s(t)| dt < \infty$$

- (h) A discrete-time LTI system is causal if and only if its step response $s(n)$ is zero for $n < 0$.

Solution

- (a) True. If $h(t)$ is periodic and nonzero, then

$$\int_{-\infty}^{\infty} |h(t)| dt = \infty$$

Therefore, $h(t)$ is unstable.

- (b) False. For example, inverse of $h(n) = \delta(n-k)$ is $g(n) = \delta(n+k)$, which is noncausal.

- (c) False. For example, $h(n) = u(n)$ implies that

$$\sum_{n=-\infty}^{\infty} |h(n)| = \sum_{n=0}^{\infty} u(n) = \sum_{n=0}^{\infty} 1 = \infty$$

This is an unstable system.

- (d) True. Assuming that $h(n)$ is bounded and nonzero in the range $n_1 \leq n \leq n_2$,

$$\sum_{n=n_1}^{n_2} |h(n)| < \infty$$

This implies that the system is stable.

- (e) False. For example, $h(t) = e^t u(t)$ is causal but unstable.

- (f) False. For example, the cascade of a causal system with impulse response $h_1(n) = \delta(n-1)$ and a noncausal system with impulse response $h_2(n) = \delta(n+1)$ leads to a system with overall impulse response $h(n) = h_1(n) * h_2(n) = \delta(n)$.

- (g) False. For example, if $h(t) = e^{-t} u(t)$, then $s(t) = \int_{-\infty}^t h(\tau) d\tau = (1 - e^{-t}) u(t)$, and

$$\begin{aligned} \int_{-\infty}^{\infty} |s(t)| dt &= \int_{-\infty}^{\infty} |(1 - e^{-t}) u(t)| dt \\ &= \int_0^{\infty} (1 - e^{-t}) dt \\ &= [t + e^{-t}]_0^{\infty} = \infty \end{aligned}$$

Although the system is stable, the step response is not absolutely integrable.

- (h) True. We may write $u(n) = \sum_{k=0}^{\infty} \delta(n-k)$. Therefore,

$$s(n) = \sum_{k=0}^{\infty} h(n-k)$$

If $s(n) = 0$ for $n < 0$, then $h(n) = 0$ for $n < 0$ and the system is causal.

Example 2.30 Find and sketch the autocorrelation function $R_{xx}(\tau)$ for

$$x(t) = e^{-at}u(t), \quad a > 0$$

Solution
By definition

$$\begin{aligned} R_{xx}(\tau) &= \int_{-\infty}^{\infty} x(t)x(t-\tau) dt \\ &= \int_{-\infty}^{\infty} e^{-at}u(t)e^{-a(t-\tau)}u(t-\tau) dt \\ &= e^{a\tau} \int_{-\infty}^{\infty} e^{-2at}u(t)u(t-\tau) dt \end{aligned}$$

For $\tau > 0$,

$$\begin{aligned} u(t)u(t-\tau) &= \begin{cases} 1, & t > \tau \\ 0, & t < \tau \end{cases} \\ R_{xx}(\tau) &= e^{a\tau} \int_{\tau}^{\infty} e^{-2at} dt \\ &= \frac{1}{2a} e^{-a\tau} \end{aligned}$$

Since $R_{xx}(\tau)$ is an even function of τ , we conclude that

$$R_{xx}(\tau) = \frac{1}{2a} e^{-a|\tau|}, \quad a > 0$$

which is sketched in Fig. 2.46.

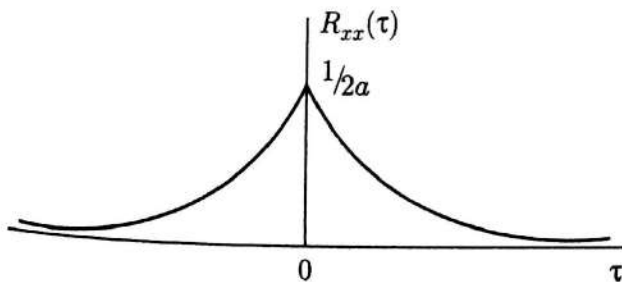


Fig. 2.46

Example 2.31 Find the autocorrelation function $R_{xx}(\tau)$ of sine wave signal

$$x(t) = A \sin(\omega_0 t + \phi), \quad \omega_0 = \frac{2\pi}{T}$$

Solution
By definition

$$\begin{aligned} R_{xx}(\tau) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t-\tau) dt \\ &= \lim_{T \rightarrow \infty} \frac{A^2}{T} \int_{-T/2}^{T/2} \sin(\omega_0 t + \phi) \sin(\omega_0(t-\tau) + \phi) dt \\ &= \lim_{T \rightarrow \infty} \frac{A^2}{2T} \int_{-T/2}^{T/2} [\cos(\omega_0 \tau) - \cos(2\omega_0 t + 2\phi - \omega_0 \tau)] dt \\ &= \lim_{T \rightarrow \infty} \frac{A^2}{2T} \int_{-T/2}^{T/2} \cos(\omega_0 \tau) dt - \lim_{T \rightarrow \infty} \frac{A^2}{2T} \int_{-T/2}^{T/2} \cos(2\omega_0 t + 2\phi - \omega_0 \tau) dt \\ &= \frac{A^2}{2T} \cos(\omega_0 \tau) \int_{-T/2}^{T/2} dt - 0 \\ &= \frac{A^2}{2} \cos(\omega_0 \tau) \end{aligned}$$

Now at $\tau = 0$,

$$R_{xx}(\tau) \Big|_{\tau=0} = R_{xx}(0) = P_x = \frac{A^2}{2}$$

Example 2.32 Determine the overall impulse response of the system shown in Fig. 2.47. Given that

$$h_1(n) = \delta(n) - a\delta(n-1)$$

$$h_2(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$h_3(n) = a^n u(n)$$

$$h_4(n) = (n-1)u(n)$$

$$h_5(n) = \delta(n) + nu(n-1) + \delta(n-2)$$

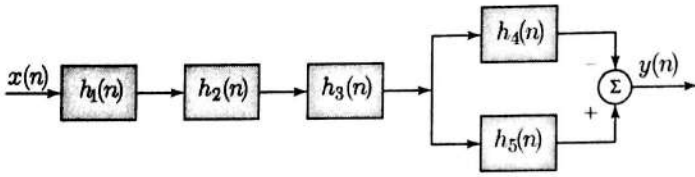


Fig. 2.47

Solution

It is clear from the figure that

$$h(n) = h_1(n) * h_2(n) * h_3(n) * [h_5(n) - h_4(n)]$$

To evaluate $h(n)$, we first perform the convolution $h_1(n) * h_3(n)$ as

$$\begin{aligned} h_1(n) * h_3(n) &= [\delta(n) - a\delta(n-1)] * a^n u(n) \\ &= a^n u(n) - a^n u(n-1) \\ &= a^n [u(n) - u(n-1)] \\ &= a^n \delta(n) = \delta(n) \end{aligned}$$

Also,

$$\begin{aligned} h_5(n) - h_4(n) &= \delta(n) + nu(n-1) + \delta(n-2) \\ &\quad - (n-1)u(n) \\ &= \delta(n) + \delta(n-2) + u(n) \end{aligned}$$

so that

$$\begin{aligned} h(n) &= \delta(n) * h_2(n) * [\delta(n) + \delta(n-2) + u(n)] \\ &= h_2(n) + h_2(n-2) + s_2(n) \end{aligned}$$

where $s_2(n)$ represents the step response corresponding to $h_2(n)$; i.e.,

$$\begin{aligned} s_2(n) &= \left(\frac{1}{2}\right)^n u(n) * u(n) \\ &= \sum_{k=0}^n \left(\frac{1}{2}\right)^k, \quad n \geq 0 \\ &= \frac{1 - (1/2)^{n+1}}{1 - (1/2)} \\ &= 2 \left[1 - \left(\frac{1}{2}\right)^{n+1} \right], \quad n \geq 0 \end{aligned}$$

$$s_2(n) = 2u(n) - \left(\frac{1}{2}\right)^n u(n)$$

Thus,

$$\begin{aligned} h(n) &= \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{2}\right)^{n-2} u(n-2) \\ &\quad + 2u(n) - \left(\frac{1}{2}\right)^n u(n) \\ &= \left(\frac{1}{2}\right)^{n-2} u(n-2) + 2u(n) \end{aligned}$$

Example 2.33 A system is described by the difference equation $y(n] = x(n] + ay(n-1]$. Find the impulse response of the inverse of this system. From the impulse response, find the difference equation of the inverse system. Assume the system is initially relaxed.

Solution

First we find the impulse response of this system by setting $x(n] = \delta(n]$ such that $y(n]$ corresponds to the impulse response $h(n]$. Thus, we may write

$$\begin{aligned} h(n) &= \delta(n) + ah(n-1) \\ h(0) &= 1 + ah(-1) = 1 + 0 = 1 \\ h(1) &= 0 + ah(0) = a \\ h(2) &= 0 + ah(1) = a^2 \\ h(3) &= 0 + ah(2) = a^3 \\ &\vdots \\ h(n) &= a^n \end{aligned}$$

Therefore,

$$h(n) = a^n u(n)$$

The inverse system must satisfy

$$h(n) * h_I(n) = \delta(n)$$

Note that

$$h(n) = \delta(n) + ah(n-1)$$

$$h(n) - ah(n-1) = \delta(n)$$

$$h(n) * [\delta(n) - a\delta(n-1)] = \delta(n)$$

comparing with

$$h(n) * h_I(n) = \delta(n)$$

we get

$$h_I(n) = \delta(n) - a\delta(n-1)$$

The input to the inverse system is $y(n)$ and output is $x(n)$.

$$\begin{aligned} x(n) &= h_I(n) * y(n) \\ &= [\delta(n) - a\delta(n-1)] * y(n) \\ &= y(n) - ay(n-1) \end{aligned}$$

Example 2.34 A signal $x(n)$ with known autocorrelation $R_{xx}(m)$ is applied to an LTI system with impulse response $h(n)$, producing the output signal

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

(a) Find the expression for $R_{yx}(m)$ in terms of $R_{xx}(m)$.

(b) Find the expression for $R_{xy}(m)$ in terms of $R_{xx}(m)$.

(c) Find the expression for $R_{yy}(m)$ in terms of $R_{xx}(m)$.

Solution

(a) The crosscorrelation between output and the input signal is

$$\begin{aligned} R_{yx}(m) &= y(m) * x(-m) \\ &= [x(m) * h(m)] * x(-m) \\ &= [x(m) * x(-m)] * h(m) \\ &= R_{xx}(m) * h(m) \end{aligned}$$

Hence the crosscorrelation between the input and the output of the system is the convolution of the impulse response with the autocorrelation of the input sequence. Alternatively, $R_{yx}(m)$ may be viewed as the output of the LTI system when the input sequence is $R_{xx}(m)$.

(b) The crosscorrelation between the input and the output signal is

$$\begin{aligned} R_{xy}(m) &= x(m) * y(-m) \\ &= x(m) * [h(-m) * x(-m)] \\ &= [x(m) * x(-m)] * h(-m) \\ &= R_{xx}(m) * h(-m) \end{aligned}$$

(c) The autocorrelation of the output signal is

$$\begin{aligned} R_{yy}(m) &= y(m) * y(-m) \\ &= [x(m) * h(m)] * [x(-m) * h(-m)] \\ &= [x(m) * x(-m)] * [h(m) * h(-m)] \\ &= R_{xx}(m) * R_{hh}(-m) \end{aligned}$$

The autocorrelation $R_{hh}(m)$ of the impulse response $h(n)$ exists if the system is stable.

SUMMARY

1. A linear time-invariant (LTI) system is completely characterized by its impulse response $h(t)$.
2. The output $y(t)$ of an LTI system is the convolution of the input $x(t)$ with the impulse response of the system

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau$$

3. The convolution operation gives only the zero-state response (i.e., forced response) of the system.

4. The convolution operator is commutative, distributive, and associative.
5. The step response of a continuous-time linear system with impulse response $h(t)$ is

$$s(t) = \int_{-\infty}^t h(\tau) d\tau$$

6. The step response of a discrete-time linear system with impulse response $h(n)$ is

$$s(n) = \sum_{k=-\infty}^n h(k)$$

7. A continuous-time LTI system is causal if $h(t) = 0$ for $t < 0$.
8. A discrete-time LTI system is causal if $h(n) = 0$ for $n < 0$.

9. A continuous-time LTI system is stable if and only if

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

10. A discrete-time LTI system is stable if and only if

$$\sum_{n=-\infty}^{\infty} h(n) < \infty$$

11. For energy signals, there is a simple mathematical relationship between correlation and convolution sum:

$$R_{xy}(m) = x(m) * y(-m)$$

12. Correlation is used to measure the similarity between the two signals.

MULTIPLE-CHOICE QUESTIONS

- A continuous-time periodic signal $x(t)$, having a period T , is convolved with itself. The resulting signal is
 - not periodic
 - periodic having a period T
 - periodic having a period $2T$
 - periodic having a period $T/2$
- The impulse response of a system described by the differential equation $d^2y(t)/dt^2 + y(t) = x(t)$ will be
 - a constant
 - an impulse function
 - a sinusoid
 - an exponentially decaying function
- The condition $\int_{-\infty}^{\infty} |h(t)| dt < \infty$ must be satisfied by a system that is
 - memoryless
 - BIBO stable
 - causal
 - invertible.
- For a system with input $x(n) = \delta(n-1)$ and impulse response $h(n) = \delta(n+1)$, the output is
 - $\delta(n-1)$
 - $\delta(n-2)$
 - $\delta(n)$
 - $\delta(n+2)$
- The unit step response of an LTI system with impulse response $h(n) = \delta(n) - \delta(n-1)$ is
 - $\delta(n-1)$
 - $\delta(n)$
 - $u(n-1)$
 - $u(n)$
- The convolution of a finite sequence with an infinite sequence
 - may be a finite or infinite sequence
 - is always a finite sequence
 - is always an infinite sequence
 - cannot be found
- Convolution is used to find
 - the impulse response of an LTI system
 - frequency response of a system
 - the time response of an LTI system
 - the phase response of an LTI system
- The step response of an LTI system when the impulse response $h(n)$ is unit step $u(n)$ is
 - $(n+1)u(n)$
 - $nu(n)$
 - $(n-1)u(n)$
 - $n^2u(n)$
- The convolution of a rectangle pulse with itself is

- (a) another rectangle pulse
(b) square pulse
(c) triangular pulse
(d) sinc pulse
10. The autocorrelation of a rectangular pulse is
(a) another rectangle pulse
(b) square pulse
(c) triangular pulse
(d) sinc pulse
11. The impulse response of a system is $h(n) = \alpha^n u(n)$. The condition for the system to be BIBO stable is
(a) α is real and positive
(b) α is real and negative
(c) $|\alpha| > 1$
(d) $|\alpha| < 1$
12. The unit impulse response of an LTI system is the unit step function $u(t)$. For $t > 0$, the response of the system to an excitation $e^{-at}u(t)$, $a > 0$, will be
(a) ae^{-at}
(b) $\frac{1 - e^{-at}}{a}$
(c) $a(1 - e^{-at})$
(d) $1 - e^{-at}$
13. When the impulse response $h(t)$ of an LTI system is integrated over the whole time domain, it gives
(a) steady-state response of the system
(b) dc response of the system
(c) transient response of the system
(d) steady-state error of the system
14. The convolution of $u(n)$ with $u(n - 4)$ at $n = 5$ is
(a) 5
(b) 2
(c) 1
(d) 0
15. A first-order all-pass filter impulse response is given by $h(t) = -\delta(t) + 2e^{-t}u(t)$. The zero state response of this filter for the input $e^t u(-t)$ is
(a) $e^{-t}u(t)$
(b) $e^{-t}u(t) + 2e^{-t}u(-t)$
(c) $e^{-2t}u(t)$
(d) $e^{-t}u(-t)$

PROBLEMS

2.1 Find the convolution $y(n) = x(n) * h(n)$ of the following signals:

(a)

$$x(n) = \begin{cases} -1, & -5 \leq n \leq -1 \\ 1, & 0 \leq n \leq 4 \end{cases}$$

$$h(n) = 2u(n)$$

(b)

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$h(n) = \delta(n) + \delta(n - 1) + \left(\frac{1}{3}\right)^n u(n)$$

(c)

$$x(n) = u(n)$$

$$h(n) = 1, \quad 0 \leq n \leq 9$$

(d)

$$x(n) = \left(\frac{1}{3}\right)^n u(n)$$

$$h(n) = \delta(n) + \left(\frac{1}{3}\right)^n u(n)$$

2.2 Determine the impulse response for the cascade of two LTI systems having impulse responses

$$h_1(n) = \left(\frac{1}{2}\right)^n u(n)$$

and

$$h_2(n) = \left(\frac{1}{4}\right)^n u(n)$$

2.3 Determine the range of values of the parameter α for which the LTI system with impulse response

$$h(n) = \alpha^n u(n)$$

is stable.

2.4 Determine the range of values of α and β for which the LTI system with impulse response

$$h(n) = \begin{cases} \alpha^n, & n \geq 0 \\ \beta^n, & n < 0 \end{cases}$$

is stable.

2.5 Compute the autocorrelation of the signal

$$x(n) = a^n u(n), \quad 0 < a < 1$$

2.6 Determine and sketch the convolution $y(n)$ of the signals

$$x(n) = \begin{cases} \frac{1}{3}n, & 0 \leq n \leq 6 \\ 0, & \text{elsewhere} \end{cases}$$

$$h(n) = \begin{cases} 1, & -2 \leq n \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

- (a) Graphically
(b) Analytically

2.7 The discrete-time system

$$y(n] = ny(n-1) + x(n)$$

is at rest [i.e., $y(-1) = 0$]. Check if the system is LTI and BIBO stable.

2.8 Find the convolution $y(n) = x(n) * h(n)$ for each of the following pairs of finite sequences:

$$(a) \quad x(n) = \left\{ 1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16} \right\}$$

$$h(n) = \{1, -1, 1, -1\}$$

$$(b) \quad x(n) = \{1, 2, 3, 0, -1\}$$

$$h(n) = \{2, -1, 3, 1, -2\}$$

$$(c) \quad x(n) = \left\{ 3, \frac{1}{2}, -\frac{1}{4}, 1, 4 \right\}$$

$$h(n) = \left\{ 2, -1, \frac{1}{2}, -\frac{1}{2} \right\}$$

where the arrow indicates the value for $n = 0$. In this notation, it is assumed that not all values listed are zero.

2.9 Let

$$x(n) = \delta(n) + 2\delta(n-1) - \delta(n-3)$$

and

$$h(n) = 2\delta(n+1) + 2\delta(n-1)$$

Compute and plot each of the following convolutions:

$$(a) \quad y_1(n) = x(n) * h(n)$$

$$(b) \quad y_2(n) = x(n+2) * h(n)$$

$$(c) \quad y_3(n) = x(n) * h(n+2)$$

2.10 Consider an input $x(n]$ and a unit impulse response $h(n]$ given by

$$x(n) = \left(\frac{1}{2}\right)^{n-2} u(n-2)$$

$$h(n) = u(n+2)$$

Determine and plot the output $y(n) = x(n) * h(n)$.

2.11 (a) Find the impulse response of the system shown in Fig. 2.48. Assume that

$$h_1(n) = h_2(n) = \left(\frac{1}{3}\right)^n u(n)$$

$$h_3(n) = u(n)$$

$$h_4(n) = \left(\frac{1}{2}\right)^n u(n)$$

(b) Find the response of the system to the unit step input.

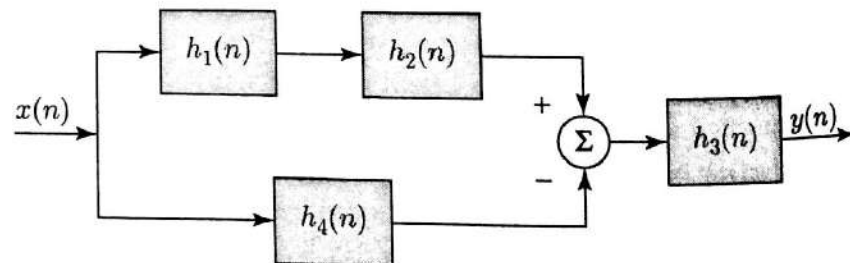


Fig. 2.48

2.12 Determine the step response of an LTI system whose impulse response is given by

$$h(n) = (-1)^n [u(n+2) - u(n-3)]$$

2.13 Let $y(t) = x(t) * h(t)$ and $g(t) = x(t/a) * h(t/a)$. Show that $g(t) = Ay(Bt)$. Determine the values of A and B .

2.14 Find the autocorrelation of the following signals:

- (a) $x(t) = e^{-at}u(t)$
 (b) $x(t) = A\Pi\left(\frac{t}{T}\right)$

2.15 Find the expression for the impulse response relating the input $x(t)$ to the output $y(t)$ in terms

of the impulse response of each subsystem for the system shown in Fig. 2.49.

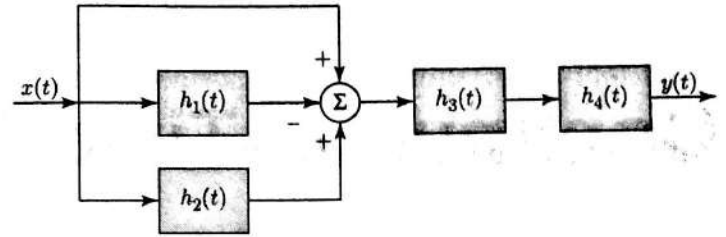


Fig. 2.49

ANSWERS TO MULTIPLE-CHOICE QUESTIONS

1. (b) 2. (c) 3. (b) 4. (c) 5. (b) 6. (c) 7. (c) 8. (a) 9. (c) 10. (c)
 11. (d) 12. (b) 13. (a) 14. (b) 15. (d)